

Time-dependent density functional theory (In progress)

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Abstract

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I. OVERVIEW AND MOTIVATION

Time-dependent density functional theory (TDDFT) is growing in importance. Diverse examples of its use and development include Refs. [1–17] as well as papers cited in surveys such as Refs. 18, 19. To date less attention has gone to the coupled motions of electrons and nuclei than to electron excitations and response with static nuclei.

Though a TDDFT for temporally periodic potentials was formulated earlier [20, 21], the now-conventional form was introduced by Runge and Gross [22], hereafter RG. As with the original ground state Hohenberg-Kohn argument [23], RG proceeded by *reductio ad absurdum* to the result that, for a v -representable time-dependent density, under certain assumptions the map between the time-dependent external potential and that density is invertible. For those assumptions, this invertibility establishes the existence, though not the structure, of a TDDFT. Issues of boundary conditions on the potentials and densities and implicit assumptions for which the RG proof is valid arose shortly [24, 25].

A Levy-Lieb constrained search [26–28] formulation of the density-potential mapping was given by Kohl and Dreizler [29]. Posed in the context of scattering problems, it did not provide a procedure for constructing solutions. Then the RG logic was extended to include time-dependent forces on spatially fixed nuclei [29–31]. However, coupled electronic and nuclear motion usually is treated via Li and Tong’s extension [32] to multi-species systems. There is a considerable variety of other work on the RG logic that, while of general relevance, is not specific to the issues treated here [33, 34].

For explicitly wave-function-based theory, a general approach to non-Born-Oppenheimer dynamics has been given by Deumens, Öhrn, and collaborators [35, 36]. They use a parameterization via group coherent states to obtain solutions to the time-dependent Schrödinger equation (TDSE). Here we take up the non-BO problem via extension of our DFT analogue [37, 38] of the Deumens-Öhrn scheme. Our formulation involves a generalization to parameterization of the TDSE by spatially varying functions (in contrast with the time-dependent complex scalars that they use).

Specifically, the parameterizing functions are the physical densities that label the density operators via equivalence classes. Density operators are emphasized in order to exploit the structure of the constrained search procedure [26, 27]. The most general form of constrained search must include both pure states and ensembles corresponding to a specified physical

density. For density operators the distinction between states and ensembles is not a barrier. Furthermore, the search is linear in the density operators but nonlinear in the states. The density operator formulation therefore provides a unified way to expose the internal structure of constrained search in forms exploitable for both analysis and approximation. See anticipatory remarks along these lines in Refs. [28] (see p. 263) and [29] (just after eq. (5)).

Ref. 37 noted that our formulation implicitly is applicable to the non-BO case. The explicit form is developed in three papers. This one gives the formal extension as a time-dependent constrained search. First we generalize Ref. 37, 38 from N -fermion operators (and corresponding Hilbert spaces) alone to explicit inclusion of the nuclear dynamics. Then we develop transformations that constitute the basis for constructive realizations of constrained search. We give two such realizations. Both exploit properties of the contraction map from the N -body density operator to the density. That map confers a density-labeled fiber bundle structure on the density operators. The kernel of that map plays a central role in parameterizing density evolution paths. The more straightforward parameterization is by reference states in each fiber plus linear combinations of elements of the kernel. The result is an explicit set of dynamical equations for TDDFT that, however, depends on a constraint which is difficult to implement. Therefore we develop an alternative parameterization of the dynamics by constructing Lie groups that map intra-fiber (i.e. amongst operators with the same density) and Lie groups that map inter-fiber (one density to another). Application of the Time-dependent Variational Principle (TDVP) provides equations of motion for the density in terms of those Lie group parameters that, in the exact case, is equivalent to the TDSE.

In the second paper, we obtain practical single-particle equations by restricting the invariance transformations to one- and two-body contributions in order to produce approximate time-dependent functionals by systematic refinement. In the limit of N -body contributions, the functionals would be exact. The results are different from the familiar TDDFT Kohn-Sham equations but play a corresponding role. By a suitable further approximation the conventional TDDFT equations are recovered in the static nucleus limit. The third paper uses the invariance transformations to formulate a path integral version of TDDFT. That provides a convenient means to treat causality issues, to analyze DFT linear response, and to treat the relationship of the stationary phase approximation to the TDVP used in the first two papers.

II. RUNGE-GROSS TDDFT - LIMITATIONS AND COUNTEREXAMPLES

Some prior results that illuminate our objectives are summarized here.

First, the Kohl-Dreizler argument [29] has limited value for coupled nuclear and electron motion because of two assumptions the electron sub-system is assumed to be in a non-degenerate ground state initially. Non-degeneracy is not a trivial requirement, while the restriction to an initial ground-state is essential to their use of a minimum principle and associated infima.

Next, motivation for the Li-Tong multi-component RG TDDFT comes from considering the original one-component RG TDDFT. For simplicity take N_n nuclei of one species and N_e electrons. RG TDDFT depends on the assumption that the external dynamic potential

$$\hat{V} = \sum_i^{N_e} v(\mathbf{r}_i, t) \quad (1)$$

has a temporal Taylor series expansion about the initial time $t = t_0$. Let the initial spatial density be $n(\mathbf{r}, t = t_0) \equiv n_0(\mathbf{r})$. Then invertibility of the potential-density map is demonstrated under this assumption by considering the contrary, to wit, two external potentials $v'(\mathbf{r}, t)$, $v(\mathbf{r}, t)$ that yield the same n and differ by more than a function of time alone. Nonexistence of such duplicate potentials requires that a certain surface integral vanish. RG argued that it vanishes for any physically meaningful potential, though they noted that the Taylor series assumption disqualifies adiabatically switched-on potentials. van Leeuwen provided an extension of that original argument, which, however, does not address the issues raised below [39].

Xu and Rajagopal gave counterexamples in which the surface integral is not zero [24]. Though Dhara and Ghosh [25] concluded that those counterexamples might not be physically realizable, they also pointed out that the RG proof is not valid for two important categories of systems, the homogeneous electron gas [?] and systems with moving bare nuclei. For the latter, the temporal Taylor series has singularities of arbitrarily high order [?]. The coefficients of the highly singular terms are time derivatives of $X(t)$ evaluated at t_0 . Those all vanish for stationary nuclei and the most singular terms in the Taylor series in that case go as the ordinary nuclear-electron potential. This problem is ignored in Ref. 3 [at their equation (12)] but see below.

For the surface integral issue, Gross and Kohn introduced two requirements on the RG

scheme, one physical, the other formal [?]. The formal constraint concerns the asymptotic behavior, with respect to r , of the Taylor series coefficients of $v(\mathbf{r}_i, t)$ and is not relevant to the present discussion. The physical requirement is that “experimentally realizable potentials are always produced by *real* (this means, in particular, normalizable) external charge densities $\rho_{ext}(\mathbf{r}t) \dots$ ” Clearly this requirement does not address the case of moving bare nuclei as the external charge densities, even though such delta-function charges are normalizable in the usual sense. It might be argued that non-singular distributions were intended. For nuclear motion this means screening, in practice, a pseudopotential. This is what Refs. 1? do in the BO approximation. Refs. 3, 16 do the same in practice, hence avoid the error just noted. Non-singular distributions are fine practically, but do not solve the basic problem. Restriction to such distributions would force TDDFT with nuclear motion never to be all-electron. In turn that would lead to the odd outcome that the time-independent limit of TDDFT would not recover normal all-electron time-independent DFT (because of the mandatory nuclear screening).

Thus one arrives at treatment of the nuclei plus electrons via the Li-Tong [32] multi-component plasma version of RG TDDFT. No full DFT dynamics (i.e., non-Kohn-Sham, even if purely formal) for the Li-Tong scheme has, to our knowledge, ever been given. Though in principle general, the nuclear motion in the Li-Tong approach in practice is purely at the KS level. See for example Gross, Dobson, and Petersilka [18]. Since KS dynamics is for the *model* nuclear system, there is a challenge to demonstrate correspondence to the dynamics of the *physical* system. In the classical limit, Gross et al. obtain the correspondence from the Ehrenfest theorem, with forces on the nuclei from gradients of the TDKS potentials. Unexplored, as far as we are aware, is the correspondence for either narrow wavepackets or full-quantum nuclei.[?] The difficulty in analyzing the dynamics lies in the invertibility arguments, which provide no insight into the structure of the space of time-dependent densities. Note also that, by construction, the Li-Tong scheme has the same dependence on convergence of temporal Taylor series expansions as the original RG scheme, hence cannot include adiabatically switched external potentials.

Keep in mind that the development given below does not appeal to the existence of a 1-1 map between external potentials and allowable densities as a route to functionals of the density. Instead our analysis proceeds by classification of states according to densities, just as in constrained search for stationary states. We then use the densities as parameters to

label states in the spirit of coherent state theory. The result is functionals of the density that have a unique value associated to each of these classes of states for all times. Equations of motion for the density follow from the TDVP.[?]

III. DENSITY MAPPINGS AND FIBER BUNDLES

The objective of this section is to show how the densities at any time can be used to classify density operators into a fiber bundle structure and to develop exploitable properties of the bundle. The benefit of this approach shows up in the following sections, where constrained search as a function of time is converted into well-characterized, explicit transformations along and among fibers. Systematic approximation of those transformations then provides a constructive route to approximate functionals, the topic of paper II.

Without loss of generality we simplify to a single nuclear species (except when multi-species aspects need to be noted). Let the generic electron and nuclear coordinates be $\mathbf{r}$ and $\mathbf{R}$ respectively and similarly s, S for the spins. The composite coordinates then will be written as

$$\begin{aligned}\mathbf{x} &\equiv (\mathbf{r}, s) \\ \mathbf{X} &\equiv (\mathbf{R}, S) \\ \mathbf{z} &\equiv (\mathbf{x}, \mathbf{X})\end{aligned}\tag{2}$$

The manifolds of N_e electronic coordinates (space and spin) and N_n nuclei will be denoted as

$$\begin{aligned}\mathbf{x} &= \{\mathbf{x}_1, \mathbf{x}_2, \dots \mathbf{x}_{N_e}\} \\ \mathbf{X} &= \{\mathbf{X}_1, \mathbf{X}_2, \dots \mathbf{X}_{N_n}\}\end{aligned}\tag{3}$$

To treat the electronic and nuclear degrees of freedom in a balanced fashion and, as well, avoid problems inherent in a formulation that decouples electron space and spin coordinates from the outset (e.g. symmetry breakings, spin contamination [? ? ?]), we work with the full space-spin electron and nuclear density

$$\gamma(\mathbf{x}, \mathbf{X}, t) \equiv \gamma(\mathbf{z}, t)\tag{4}$$

rather than the more familiar spatial density $n(\mathbf{r}, \mathbf{R}, t)$ or spin-polarized spatial densities. Note that γ integrates to the product $N_e N_n$, not their sum. $\gamma(\mathbf{x}, \mathbf{X}, t)$ can be interpreted as

the joint probability for the simultaneous occurrence at t of an electron at position $\mathbf{x}$ and a nucleus at position $\mathbf{X}$. BWBW This has the further benefit of avoiding possible problems introduced by the translational invariance of isolated systems [?].BWBW The single species space-spin densities can be recovered by integration

$$\begin{aligned}\gamma_e(\mathbf{x}, t) &= \frac{1}{N_n} \int \gamma(\mathbf{x}, \mathbf{X}, t) d\mathbf{X} \\ \gamma_n(\mathbf{X}, t) &= \frac{1}{N_e} \int \gamma(\mathbf{x}, \mathbf{X}, t) d\mathbf{x}\end{aligned}\tag{5}$$

With two species, it is easiest to formulate the problem in second quantization. The field operators are

$$\begin{aligned}\Phi_e^\dagger(\mathbf{x}) &= \sum_i \phi_i^*(\mathbf{x}) a_i^\dagger \\ \Phi_n^\dagger(\mathbf{X}) &= \sum_j \phi_{n,j}^*(\mathbf{X}) b_j^\dagger\end{aligned}\tag{6}$$

where the $\phi_i(\mathbf{x})$, $\phi_{n,j}(\mathbf{X})$ are elements of complete one-particle basis sets for electrons and nuclei respectively and $a_i^\dagger$, $b_j^\dagger$ are electron and nuclear creation operators respectively, with commutation relations

$$\begin{aligned}[a_i^\dagger, a_j]_+ &= \delta_{ij} \\ [a_i^\dagger, a_j^\dagger]_+ &= [a_i, a_j]_+ = 0 \\ [b_i^\dagger, b_j]_\pm &= \delta_{ij} \\ [b_i^\dagger, b_j^\dagger]_\pm &= [b_i, b_j]_\pm = 0\end{aligned}\tag{7}$$

Commutation or anti-commutation depends on the nuclear species (of course) and all inter-species commutators are zero. Frequently products of these operators for both an electron and a nucleus occur. A compressed notation for such products

$$\Phi^\dagger(\mathbf{z})\Phi(\mathbf{z}) \equiv \Phi_e^\dagger(\mathbf{x})\Phi_e(\mathbf{x})\Phi_n^\dagger(\mathbf{X})\Phi_n(\mathbf{X})\tag{8}$$

will be used.

Generalizing Refs. 37, 38, the total system density operator $D^N(t)$ is defined in the tensor product Hilbert space of the N_e electrons and N_n nuclei, with $N = N_e + N_n$. The time-dependent density then is

$$\gamma(\mathbf{z}, t) = Tr \{ \Phi^\dagger(\mathbf{z})\Phi(\mathbf{z})D^N(t) \} = \gamma(\mathbf{x}, \mathbf{X}, t)\tag{9}$$

In the coordinate representation, the total system wave function $\Psi(\mathbf{x}, \mathbf{X}, t)$ can be expressed in terms of the vacuum state $|0\rangle$ as

$$\begin{aligned}\Psi(\mathbf{x}, \mathbf{X}, t) &= \langle \mathbf{x}_1, \dots, \mathbf{x}_{N_e}, \mathbf{X}_1, \dots, \mathbf{X}_{N_n} | \Psi(t) \rangle \\ &= \langle 0 | \Phi_e(\mathbf{x}_1) \dots \Phi_e(\mathbf{x}_{N_e}) \Phi_n(\mathbf{X}_1) \dots \Phi_n(\mathbf{X}_{N_n}) \Psi(t) \rangle\end{aligned}\quad (10)$$

where $|\Psi(t)\rangle \in \mathcal{H}^{N_e} \otimes \mathcal{H}^{N_n}$ and the associated N -body density operator integral kernels are

$$\begin{aligned}D^N(\mathbf{x}'_1, \dots, \mathbf{x}'_{N_e}, \mathbf{X}'_1, \dots, \mathbf{X}'_{N_n}; \mathbf{x}_1, \dots, \mathbf{x}_{N_e}, \mathbf{X}_1, \dots, \mathbf{X}_{N_n}, t) \\ = \Psi^*(\mathbf{x}'_1, \dots, \mathbf{x}'_{N_e}, \mathbf{X}'_1, \dots, \mathbf{X}'_{N_n}, t) \Psi(\mathbf{x}_1, \dots, \mathbf{x}_{N_e}, \mathbf{X}_1, \dots, \mathbf{X}_{N_n}, t)\end{aligned}\quad (11)$$

(Remark that these objects are called density matrices in the theoretical chemistry literature, terminology that is inconvenient here.) The one-electron density operator kernel $D_e^1(\mathbf{x}', \mathbf{x}, t)$ arises from integrating out all the other electron coordinates *and all* the nuclear coordinates. The converse holds for the nuclear density operator kernel. In the BO approximation, the electron density operator kernel is a conditional probability relative to the nuclear positions. A relevant discussion is in Sjöqvist and Hedström [?].

With one substantive alteration, the previous formulation in Refs. 37, 38, involving trace class operators, Hilbert-Schmidt operators, and contraction maps to one-particle function spaces goes through essentially unchanged (except for obvious nuclear spin symmetry details). The alteration arises because it is useful to formulate the TDVP in terms of unnormalized states and handle the normalization explicitly.

The central quantity is the contraction map that yields the combined density from the density operators

$$\begin{aligned}\Xi : D^N(t) &\rightarrow \gamma(\mathbf{z}, t) \\ \gamma(\mathbf{z}, t) &= \text{Tr}\{\Phi^\dagger(\mathbf{z})\Phi(\mathbf{z})D^N(t)\}\end{aligned}\quad (12)$$

Here

$$\begin{aligned}\text{Tr}\{D^N(t)\} &= 1 \quad \forall t \\ \int \gamma(\mathbf{z}, t) d\mathbf{z} &= N_e N_n \quad \forall t\end{aligned}\quad (13)$$

Ξ itself is a linear map from the space $\mathcal{B}_1(\mathcal{H}^N)$ of trace class operators on N -particle Hilbert space to absolute integrable functions on $\mathbb{R}^6 \times \mathbb{C}^4$.

The density labels equivalence classes $[\gamma]_N$ of positive trace-class operators Y and W by

$$Y, W \in [\gamma]_N \Leftrightarrow \Xi(Y) = \Xi(W) = \gamma \quad (14)$$

These equivalence classes are elements of a bundle [?]. By inspection it is a fiber bundle (specifically an affine bundle) [?]; see below. Denote it as $\mathfrak{B}_{aff}$. The base of the bundle is the set of densities, which belong to $L_1(\mathbb{R}^6 \times \mathbb{C}^4)$. The elements of a fiber are the members of an equivalence class which form an affine space; see below. The identification follows.

If restricted to normalized, positive, trace-class operators, this classification is physically equivalent to the original constrained search scheme states are grouped by the density they yield. There are analytical advantages to working with the density operators instead of the states (the problem is linear in the density operators but nonlinear in the states). To gain those advantages it helps to extend consideration, temporarily, to all trace class operators. Doing so has two consequences that must be addressed eventually (i) not all the quantities f generated by the contraction map Ξ are physical densities γ (everywhere positive), (ii) for a genuine physical density γ not every trace-class operator that contracts to it is positive. With these caveats in mind, the extended equivalence class is

$$[f]_N = \{Y \in \mathcal{B}_1(\mathcal{H}^N); \Xi(Y) = f\} \quad (15)$$

(Aside different f 's do not necessarily have the same norm.) The advantage gained is that the kernel of Ξ

$$\ker\{\Xi\} = \{Y; Tr[\Phi^\dagger \Phi Y] = 0\} \equiv \{U - W; \Xi(U) = \Xi(W)\} \quad (16)$$

is a linear subspace of $\mathcal{B}_1(\mathcal{H}^N)$. Then every element in a given equivalence class is expressible as

$$Y = \sum_{i=1}^d \alpha_i W_i \quad ; \quad \sum_{i=1}^d \alpha_i = 1 \quad (17)$$

BWBW which shows that $[f]_N$ is an affine space, and BWBW where

$$W_i = Y_{ref} + \xi_i; \quad i > 0 \quad (18)$$

Here Y_{ref} is a fixed, arbitrary element of the particular $[f]_N$ (i.e. is not in $\ker\{\Xi\}$) and the $\{\xi_i\}$ constitute a basis for $\ker \Xi$ [?]. Each fiber thus is identified by a reference element in

$[f]_N$, while its internal structure is characterized by the kernel via eqs. (17, 18). Eq. (17) can be recast as

$$Y = Y_{ref} + U$$

$$U = \sum_{i=1}^d \alpha_i \xi_i \quad (19)$$

and the bundle is affine [?]. It follows that the orthogonal projector onto any pure N -particle state $|\Psi_N\rangle$ with density γ can be written as

$$\frac{|\Psi_N(\mathbf{c})\rangle \langle \Psi_N(\mathbf{c})|}{\langle \Psi_N(\mathbf{c}) | \Psi_N(\mathbf{c}) \rangle} = Y_{ref}(\gamma) + \sum_{i=1}^d c_i \xi_i \quad (20)$$

This expression is key to pure-state constrained search. Changes in values of the $\{c_i\}$ move within a fiber, i.e. search over all system density operators corresponding to a fixed density. Changing to another fiber means another density has been selected. Both TDDFT and ground state DFT result by combination with the appropriate variation principle (stationary action for TDDFT, Rayleigh-Ritz variation principle for ground state).

IV. EXPLICIT IMPLEMENTATION OF TIME-DEPENDENT CONSTRAINED SEARCH

The next task is to implement the search along and between fibers at any time to determine the solution of the time-dependent Schrödinger equation (TDSE). Refs. 37, 38 extended the work of Kramer and Sarceno [?] (“K&S” to distinguish from Kohn and Sham, commonly denoted “KS”) to include parameterization of the TDSE via the density as a function of time and electronic coordinate $\mathbf{x}$. This section gives the extension to the density for coupled electron and nuclei as functions of the coordinate $\mathbf{z}$ and time t and the resulting dynamics in terms of the coefficients $\mathbf{c}$ in eq. (20) and the density γ . To exploit the fiber bundle properties the argument here differs a bit from that of Ref. 37.

A straightforward extension of the original K&S analysis of the TDSE can be made for the manifold of states $|\Psi(\lambda(t))\rangle$ parameterized by real, time-dependent functions $\lambda(t, \mathbf{z})$. For these states, the solutions of the TDSE, denoted $|\Psi(\lambda_{\#}(t))\rangle$, are characterized by parameterizing functions that satisfy

$$\int \eta(t, \mathbf{y}, \mathbf{y}') \frac{d\lambda(\mathbf{y}')}{dt} d\mathbf{y}' = \frac{\delta \mathcal{E}(t)}{\delta \lambda(\mathbf{y})} \quad \forall \mathbf{y} \quad (21)$$

where

$$\eta(t, \mathbf{y}, \mathbf{y}') = i \left\{ \frac{\delta}{\delta \tilde{\lambda}(\mathbf{y})} \frac{\delta}{\delta \lambda(\mathbf{y}')} - \frac{\delta}{\delta \lambda(\mathbf{y})} \frac{\delta}{\delta \tilde{\lambda}(\mathbf{y}')} \right\} \ln \left\langle \Psi(\tilde{\lambda}(t)) | \Psi(\lambda(t)) \right\rangle \big|_{\tilde{\lambda}=\lambda} \quad (22)$$

and

$$\mathcal{E}(t) = \frac{\langle \Psi(\lambda(t)) | H | \Psi(\lambda(t)) \rangle}{\langle \Psi(\lambda(t)) | \Psi(\lambda(t)) \rangle} \quad (23)$$

Note the explicit norm. The metric (in the space of parameterizing functions) satisfies

$$\eta(t, \mathbf{y}, \mathbf{y}') = -\eta(t, \mathbf{y}', \mathbf{y}) \quad (24)$$

Eq. (21) can be cast in a form analogous with classical mechanics, to wit

$$\begin{aligned} \frac{d\lambda(\mathbf{y}')}{dt} &= \int \eta(t, \mathbf{y}', \mathbf{y})^{-1} \frac{\delta \mathcal{E}(t)}{\delta \lambda(\mathbf{y})} d\mathbf{y} \quad \forall \mathbf{y}' \\ \dot{\lambda}(\mathbf{y}') &= \{\lambda(\mathbf{y}), \mathcal{E}\} \end{aligned} \quad (25)$$

with the generalized Poisson bracket

$$\{f(t), g(t)\} \equiv \int \frac{\delta f(t)}{\delta \lambda(\mathbf{y})} \zeta(t, \mathbf{y}, \mathbf{y}') \frac{\delta g(t)}{\delta \lambda(\mathbf{y}')} d\mathbf{y} d\mathbf{y}' \quad (26)$$

and the definition

$$\zeta(t, \mathbf{y}, \mathbf{y}') \equiv \eta^{-1}(t, \mathbf{y}, \mathbf{y}') \quad (27)$$

all under the assumption that η^{-1} exists. This is not a strongly restrictive assumption [?].

If the manifold of states $|\Psi(\lambda(t))\rangle$ is all of $\mathcal{H}^N$, then solutions of Eqs. (25), (26) are equivalent to solutions of the full TDSE except for an overall phase that also can be determined within the parameterization [? ?]. It follows from eq. (20) that, indeed

$$\mathcal{H}^N = \{|\Psi_N(\mathbf{c}, \gamma)\rangle\} \quad (28)$$

with an obvious change in notation to account for the γ dependence. Note that the set of coefficients $\mathbf{c}$ must be constrained such that the result in eq. (20) is a pure-state projector, in symbols

$$\{c_i\} \subset \Lambda_{Y_{ref}(\gamma)} \ni Y = \text{pure state density operator} \quad (29)$$

The problem of characterizing the set Λ (or finding an alternative) is deferred to the next section. Proceeding, the equation of motion (25) that is fully equivalent to the TDSE becomes

$$\{(\mathbf{c}, \gamma), \mathcal{E}\} = (\dot{\mathbf{c}}, \dot{\gamma}) \quad (30)$$

Since all the quantities involved are functionals of γ , this equation demonstrates the existence of a TDDFT independent of the RG logic. The claim becomes evident upon partitioning eq. (30) between the $\mathbf{c}(t)$ and $\gamma(t)$. Details are in Appendix A. The outcome is that the density dynamics can be displayed explicitly, at least formally, with the c_i as functionals of γ . In terms of them, the exact temporal path for the density is the solution of

$$\dot{\gamma}(t) = \tilde{\mathcal{N}}_1(\gamma(t)) \left[\nabla_{\gamma} \mathcal{E}(\gamma(t)) - \tilde{\mathcal{N}}_2(\gamma(t)) \right] \quad (31)$$

where $\tilde{\mathcal{N}}_1(\gamma(t))$, $\tilde{\mathcal{N}}_2(\gamma(t))$ are density functionals determined by the solutions of the equation of motion for the coefficients $\mathbf{c}$, hence self-consistently. See Appendix A.

V. DYNAMICS PARAMETERIZED BY GROUP PROPERTIES OF THE FIBER BUNDLE

Parameterization via the $\mathbf{c}$ coefficients has the merit of a clear, straightforward linkage to the structure of constrained search. There seems to be no straightforward way, however, to implement the constraint eq. (29) requiring the c_i to preserve positivity of the operator Y in eq. (19). Observe that this constraint is in addition to having to select an approximation to model the states $|\Psi_N(\mathbf{c}, \gamma)\rangle$. A suggestive aspect of the $\mathbf{c}$ parameterization is that it is not complex analytic in general. A complex analytic parameterization is technically useful for dealing with the positivity constraint, as shown in this Section. Such a parameterization with independent real and imaginary parts always is equivalent to a real parameterization, hence does not violate the assumptions that give eqs. (21) - (23).

Thus we develop an alternative parameterization based on group properties of transformations of a vector bundle, $\mathfrak{B}_{vect}$, that is isomorphic to the affine bundle, $\mathfrak{B}_{aff}$, constructed in Section 3. This does lead to a complex analytic description. Use of this vector bundle also makes the group theoretic analysis less demanding because we only need to consider transformations of vector spaces rather than affine spaces. The vector bundle $\mathfrak{B}_{vect}$ is based on the direct sum decomposition of $\mathcal{B}_1(\mathcal{H}^N)$ given by

$$\mathcal{B}_1(\mathcal{H}^N) = \ker \Xi \oplus \ker \Xi^c \quad (32)$$

It has a base space $\ker \Xi^c$, the vector space complement of $\ker \Xi$ in $\mathcal{B}_1(\mathcal{H}^N)$, which is isomorphic to $L_1(\mathbb{R}^6 \times \mathbb{C}^4)$, and fibers $\ker \Xi$ that are all identical. Note therefore that the

base of $\mathfrak{B}_{vect}$ is isomorphic with the set of absolutely integrable functions that contain the set of densities and that each equivalence class $[f]_N$ is comprised of the fiber plus a particular base element. Both the fibers and the base are vector spaces.

Transformations of $\mathcal{B}_1(\mathcal{H}^N)$ that are block-triangular with respect to bases adapted to this decomposition, that is, are of the form

$$\begin{pmatrix} |\ker \Xi\rangle \langle \ker \Xi| & |\ker \Xi\rangle \langle \ker \Xi^c| \\ 0 & |\ker \Xi^c\rangle \langle \ker \Xi^c| \end{pmatrix} \quad (33)$$

induce vector bundle maps $\mathfrak{B}_{vect} \rightarrow \mathfrak{B}_{vect}$ and form the group, $\mathfrak{G}_\Xi$, of automorphisms of $\mathfrak{B}_{vect}$ [?]. Here $|A\rangle \langle B|$ denotes mappings from the subspace B to the subspace A and the lower block of eq. (33) is zero because mappings from the fiber to the base destroy the vector bundle properties.

Usually invariance groups characterizing a property of a class of states are constructed from unitary transformations. In the present case there is no unitary group of transformations that operates homogeneously, (i.e. produces all pure states from any given reference pure state), both in $\mathcal{H}^N$ and the set of pure states contained in each equivalence class $[\gamma]_N$ and also preserves the vector bundle structure of $\mathfrak{B}_{vect}$. However one can construct a commutative, but non unitary, Lie group, $\mathfrak{G}_{\ker}$, of transformations contained in the general linear group, $GL(\mathcal{H}^N)$, of $\mathcal{H}^N$ that does have these properties. This group induces positive congruence maps of $\mathcal{B}_1(\mathcal{H}^N) \mapsto \mathcal{B}_1(\mathcal{H}^N)$, hence maps states to states (see Appendix B for details). The group $\mathfrak{G}_{\ker}$ contains subgroups $\mathfrak{G}_{den}$ and $\mathfrak{G}_{fib}$ that can be used to give parameterizations of $\ker \Xi^c$ and $\ker \Xi$ respectively.

The parameterization of the base $\ker \Xi^c$ that we produce is non redundant (i.e. one set of parameters corresponds to one element and one element only) and is given in terms of the group $\mathfrak{G}_{den} \subset \mathfrak{G}_{\ker}$ which is represented by transformations of the form

$$\begin{pmatrix} t(T_B) & 0 \\ 0 & T_B \end{pmatrix} \quad (34)$$

acting on a reference element in $\ker \Xi^c$, where the top block is determined by the bottom one. In the case of the fibers, $\ker \Xi$, we produce a non redundant parameterization of *pure states* given in terms of the group $\mathfrak{G}_{fib}$ of transformations

$$\begin{pmatrix} T_F & 0 \\ 0 & I_B \end{pmatrix} \quad (35)$$

acting on a pure state reference, where the bottom block is the identity operation on the base. A more detailed discussion of the properties of these transformations, which are determined by integral kernels of operators acting in $\mathcal{H}^N$, that are diagonal in the coordinate representation, are given later in this section and in the Appendices.

States (pure or ensemble) of a physical system correspond to positive trace class operators. In the case of pure states, these operators correspond to elements of $\mathcal{H}^N$. As we are interested in mappings between states, positive maps of such operators are critical. Since all physical densities can be associated with pure states, we focus on them. Relevant details of positive operators and maps are summarized in Appendix C. In particular, the only two possibilities for mappings between pure states are unitary transformations and the more general congruence transformations[?] of $\mathcal{B}_1(\mathcal{H}^N)$. To incorporate both types, we therefore consider the most general possible invertible transformations on $\mathcal{H}^N$, namely those of the General Linear group $GL(\mathcal{H}^N)$

$$T \in GL(\mathcal{H}^N) \ni T|\Psi\rangle = |\chi\rangle \text{ and } T^{-1}|\chi\rangle = |\Psi\rangle \quad \forall |\Psi\rangle, |\chi\rangle \in \mathcal{H}^N \quad (36)$$

Because norms appear explicitly in the TDVP, the non-unitarity of these maps is not a difficulty. It follows that

$$T = e^A e^{iB} \quad (37)$$

with $A^\dagger = A$ and $B^\dagger = B$ (see Appendix C). All pure state density operators are mapped to one another by such transformations; see Appendix D on this point as well as for the basis of a systematic extension of the present analysis to ensemble density operators.

The most general group, $\mathfrak{G}_{\text{ker}} \subset \mathfrak{G}_\Xi \subset GL(\mathcal{B}_1(\mathcal{H}^N))$, of positive transformations of $\mathcal{B}_1(\mathcal{H}^N)$ that both preserve the fiber bundle structure of $\mathfrak{B}_{\text{vect}}$ and also are congruence transformations then has the following property

$$\mathfrak{G}_{\text{ker}} = \{T \in GL(\mathcal{H}^N); Tr\{\Phi^\dagger \Phi T Y T^\dagger\} = 0 \quad \forall Y \in \ker \Xi\} \quad (38)$$

By resort to the coordinate eigenfunction representation, we show in Appendices E and F that only certain subgroups of the commutative subgroup $\mathcal{C}(\mathcal{H}^N)$ of diagonal coordinate multi-particle transformations $T(\Upsilon)$ have this property. In terms of the field operators

$\{\Phi^\dagger(\kappa), \Phi(\kappa); \kappa = (\mathbf{x}, \mathbf{X})\}$, these diagonal transformations are

$$\begin{aligned}\mathcal{C}(\mathcal{H}^N) &= \left\{ T(\Upsilon); T(\Upsilon)YT^\dagger(\Upsilon) \equiv \widehat{T}(\Upsilon)(Y) ; \right. \\ T(\Upsilon) &= e^{\int \Upsilon(\kappa_1, \dots, \kappa_N) \Phi^\dagger(\kappa_1) \Phi(\kappa_1) \dots \Phi^\dagger(\kappa_N) \Phi(\kappa_N) d\kappa_1 \dots d\kappa_N} \\ \Upsilon(\kappa_1, \dots, \kappa_N) &\in \mathbb{C} \end{aligned} \quad (39)$$

Here and in the Appendices the coordinate strings are in the compressed notation

$$\kappa_1 \dots \kappa_{N_e}, \kappa_{N_e+1} \dots \kappa_N \equiv \mathbf{x}_1 \dots \mathbf{x}_{N_e}, \mathbf{X}_1 \dots \mathbf{X}_{N_n} \quad (40)$$

By construction $\mathcal{C}(\mathcal{H}^N)$ has the property

$$[T(\Upsilon), \Phi^\dagger \Phi] = 0 \quad (41)$$

Note that *all* states (unnormalized in general) belonging to $\mathcal{H}^N$ can be expressed as

$$|\Psi(\Upsilon)\rangle = T(\Upsilon)|\Psi\rangle \quad (42)$$

for any fixed reference state $|\Psi\rangle \in \mathcal{H}^N$; see Appendix H. Separation of Υ into real and imaginary parts as in eq. (37) gives

$$\begin{aligned} T(\Upsilon) &= e^{\int \mathcal{V}(\kappa_1, \dots, \kappa_N) \Phi^\dagger(\kappa_1) \Phi(\kappa_1) \dots \Phi^\dagger(\kappa_N) \Phi(\kappa_N) d\kappa_1 \dots d\kappa_N} \\ &\times e^{i \int \omega(\kappa_1, \dots, \kappa_N) \Phi^\dagger(\kappa_1) \Phi(\kappa_1) \dots \Phi^\dagger(\kappa_N) \Phi(\kappa_N) d\kappa_1 \dots d\kappa_N} \\ &\equiv e^{\check{\mathcal{V}}} e^{i\check{\omega}} \end{aligned} \quad (43)$$

The group of transformations, $\mathcal{G}_\Xi$, generated by the imaginary part of T is a unitary subgroup of $\mathcal{C}(\mathcal{H}^N)$ that leaves $\ker \Xi$ invariant

$$\begin{aligned} \mathcal{G}_\Xi &= \left\{ U(\omega); \hat{U}(\omega)(Y) \equiv U(\omega)YU^\dagger(\omega); \omega(\kappa_1, \dots, \kappa_N) \in \mathbb{R} \right. \\ U(\omega) &= e^{i \int \omega(\kappa_1, \dots, \kappa_N) \Phi^\dagger(\kappa_1) \Phi(\kappa_1) \dots \Phi^\dagger(\kappa_N) \Phi(\kappa_N) d\kappa_1 \dots d\kappa_N} \end{aligned} \quad (44)$$

Elements of this subgroup map each fiber to itself i.e. are intra fiber maps

$$Tr\{\Phi^\dagger \Phi U(\omega) Y U^\dagger(\omega)\} = Tr\{\Phi^\dagger \Phi Y\} \quad \forall Y \in \mathcal{B}_1(\mathcal{H}^N) \quad (45)$$

In fact $\mathcal{G}_\Xi$ is a subgroup of *the* invariance group of the contraction map Ξ , since

$$\Xi(U(\omega)YU^\dagger(\omega)) = \Xi(Y) \quad \forall Y \in \mathcal{B}_1(\mathcal{H}^N) \quad (46)$$

The group $\mathfrak{G}_{\ker}$ can be expressed as the direct product $\mathcal{G}_{\ker} \times \mathcal{G}_{\Xi}$, where the subgroup, $\mathcal{G}_{\ker}$, is generated by $\mathcal{V}$ (the real part of Υ).

To construct a representation of this subgroup, which contains both inter and intra fiber maps, restrictions must be imposed upon $\mathcal{V}(\kappa_1, \dots, \kappa_N)$ so that eq. (38) is satisfied. In Appendix F we show that the diagonal integral kernel T_d of this transformation may be written as

$$|T_d(\kappa_1, \dots, \kappa_N)|^2 = 1 + \phi(\kappa_1, \dots, \kappa_N) = e^{2\mathcal{V}(\kappa_1, \dots, \kappa_N)} \quad (47)$$

(note Appendix F, eq. (F9)), i.e., $T_d = e^{\check{\mathcal{V}}} = \left(1 + \check{\phi}\right)^{\frac{1}{2}}$. This decomposition is significant because of the requirement (see Appendix F) that $\check{\phi}$ needs to be strongly orthogonal to K_D , the diagonal part of $\ker \Xi$ in the coordinate representation. In other words, the integral kernel $\phi(\kappa_1, \dots, \kappa_N)$ of $\check{\phi}$ must be strongly orthogonal to the integral kernels of all diagonal operators that lie in $\ker \Xi$. This condition is obtained, (see Appendix F), by considering the decomposition of $Y \in \mathcal{B}_1(\mathcal{H}^N)$ as

$$Y = Y_{K_{OD}} + Y_{K_D} + \Gamma(\gamma) \quad (48)$$

where we decompose $\ker \Xi$ into off diagonal and diagonal parts, i.e. $\ker \Xi = K_{OD} \oplus K_D$, and $Y_{K_{OD}} \in K_{OD}$, $Y_{K_D} \in K_D$, and $\Gamma(\gamma) \in \ker \Xi^c$. Observe that the transformed diagonal part $\left(1 + \check{\phi}\right)^{\frac{1}{2}} Y_{K_D} \left(1 + \check{\phi}\right)^{\frac{1}{2}} = \left(1 + \check{\phi}\right) Y_{K_D}$ only belongs to Y_{K_D} if $\phi(\kappa_1, \dots, \kappa_N)$ is strongly orthogonal to $Y_{K_D}(\kappa_1, \dots, \kappa_N)$, while $\left(1 + \check{\phi}\right)^{\frac{1}{2}} Y_{K_{OD}} \left(1 + \check{\phi}\right)^{\frac{1}{2}\dagger}$ always belongs to K_{OD} (note that $\left(1 + \check{\phi}\right)^{\frac{1}{2}}$ does not commute with $Y_{K_{OD}}$) and $\left(1 + \check{\phi}\right) \Gamma(\gamma)$ belongs to $\ker \Xi^c$ both in the case when $\check{\phi}$ is strongly orthogonal to $\Gamma(\gamma)$ and when it is not. Clearly this map maintains the vector bundle structure of $\mathfrak{B}_{vect}$ as it does not send elements of the fibers to the base $\ker \Xi^c$.

Not all elements of $\mathcal{G}_{\ker}$ are intra-fiber maps, as in general $\left(1 + \check{\phi}\right) \Gamma(\gamma) = \Gamma(\gamma') + Y'_{K_D}$, and thus $\left(1 + \check{\phi}\right)$ is an intra-fiber map only if $\gamma' = \gamma$. In terms of eq. (19), the issue is that elements of $\mathcal{G}_{\ker}$ may change $Y_{ref} \rightarrow Y'_{ref}$, where Y'_{ref} is in a different fiber to Y_{ref} . However, $\mathcal{G}_{\ker}$ does contain a subgroup, $\mathcal{G}_{fib}$, of intra fiber maps. They are treated in Appendix F also. An essential result is that the $\check{\mathcal{V}}_{fib}$ that generate this group are expressed in terms of $\check{\phi}$ that are strongly orthogonal to K_D and act as the zero operator on S_D . Together with the group $\mathcal{G}_{\Xi}$ the group $\mathcal{G}_{fib}$ forms, via direct product, the group $\mathfrak{G}_{fib}$ of intra fiber super multiplication operators, acting in $\mathcal{B}_1(\mathcal{H}^N)$ mapping $\ker \Xi \mapsto \ker \Xi$, generated by multiplication operators

(in the coordinate representation) acting in $\mathcal{H}^N$ i.e. $\mathfrak{G}_{fib} = \mathcal{G}_{fib} \times \mathcal{G}_{\Xi}$. This group can be used to give a non redundant parameterization of $\ker \Xi$, as we show in Appendix G.

A representation of the group, $\mathfrak{G}_{den}$, of maps between fibers, that is contained in $\mathcal{G}_{ker}$, can be constructed (see Appendix F) using $\check{\mathcal{V}}_{den}$ based on $\check{\phi}$ that are strongly orthogonal to all diagonal operators that lie in $\ker \Xi$ but are not strongly orthogonal to S_D . A non redundant parameterization of S_D based on a specific (but arbitrary) reference state $D_{ref}^N(\gamma_0)$ can be formed using this representation (see Appendix G). It is possible to choose the reference state $D_{ref}^N(\gamma_0)$ and all the states generated from it, (one for each fiber), to be Independent Particle States or to be states of any prescribed form.

As a result, a non redundant parameterization of the set of pure states in $\mathfrak{B}_{vec}$ can be given in terms of the action of the product $\mathfrak{G}_{fib} \times \mathfrak{G}_{den} = \mathcal{G}_{fib} \times \mathcal{G}_{\Xi} \times \mathfrak{G}_{den} \equiv \mathfrak{G}_{pure}$ on a pure state reference element. Elements of $\mathcal{G}_{fib} \times \mathcal{G}_{\Xi}$ are intra-fiber maps, while elements of $\mathfrak{G}_{den}$ are inter-fiber maps. The state $D_{ref}^N(\mathcal{V}_{den}) \equiv D_{ref}^N(\gamma)$ obtained through the non redundant action of $\mathfrak{G}_{den}$ on $D_{ref}^N(\gamma_0)$ can be used as the reference state for the non redundant parameterization of the pure states contained in $[\gamma]_N$ through the action of $\mathfrak{G}_{fib}$ on $D_{ref}^N(\gamma)$, that gives $D^N(\mathcal{V}_{fib}(\gamma), \omega)$. Note that these maps are rank-preserving. The physical significance is that while these maps do *not* explore all elements of every fiber, they *do* explore all pure-state elements of every fiber if the initial reference is a pure state. Technical details are in Appendices F and G. In the following diagram, we collect the primary relationships described in this section. The top line is the domain, the bottom line the range of the mappings shown in the middle.

$$\begin{array}{c}
\overbrace{\mathfrak{B}_{aff} \simeq \mathfrak{B}_{vect} \simeq \mathcal{B}_1(\mathcal{H}^N) \equiv \ker \Xi \oplus \ker \Xi^c} \\
\left| \begin{array}{c}
\mathfrak{G}_\Xi \subset GL(\mathfrak{B}_1(\mathcal{H}^N)) \\
\cup \\
\mathfrak{G}_{\ker} = \mathcal{G}_{\ker} \times \mathcal{G}_\Xi \\
\cup \\
\mathfrak{G}_{pure} = \mathfrak{G}_{fib} \times \mathfrak{G}_{den} \\
\parallel \\
\mathcal{G}_{fib} \times \mathcal{G}_\Xi \times \mathfrak{G}_{den}
\end{array} \right| \\
\overbrace{\mathfrak{B}_{aff} \simeq \mathfrak{B}_{vect} \simeq \mathcal{B}_1(\mathcal{H}^N) \equiv \ker \Xi \oplus \ker \Xi^c} \\
\mathfrak{G}_{\ker} \subset \mathcal{C}(\mathcal{H}^N) \cap \text{Congruence Transforms of } \mathcal{B}_1(\mathcal{H}^N)
\end{array} \tag{49}$$

VI. EQUATIONS OF MOTION

Now we use the parameterization of pure states in $\mathcal{H}^N$ provided by the factorization of $\mathfrak{G}_{pure}$ in terms of $\mathcal{G}_\Xi$, $\mathcal{G}_{fib}$, and $\mathcal{G}_{den}$ to express the TDVP in terms of functionals of the physical density in a form amenable to sequential approximation. First a reference state $|\Psi_0\rangle \in \mathcal{H}^N$ must be selected. Such a state could be of a simple form (e.g. an Independent Particle State) that leads to straightforward expressions for expectation values. Or, it could be a correlated state (e.g., the product of an electronic Antisymmetrized Geminal Power State [?] and a variable width wave packet for nuclear positions) that provides a better basis for approximations - though at the expense of more complicated expressions for expectation values. Given such a reference state $|\Psi_0\rangle$ and its associated density γ_0 , *any* pure state (in general, unnormalized) in $\mathcal{H}^N$ can be parameterized in terms of the foregoing group factorization as

$$|\Psi(\mathcal{V}_{fib}(\gamma), \omega, \mathcal{V}_{den})\rangle = e^{\check{\mathcal{V}}_{fib}(\gamma)} e^{i\check{\omega}} e^{\check{\mathcal{V}}_{den}} |\Psi_0\rangle \tag{50}$$

where

$$\begin{aligned}
e^{\check{\mathcal{V}}_{fib}(\gamma)} &\in \mathcal{G}_{fib} \\
e^{\check{\mathcal{V}}_{den}} &\in \mathfrak{G}_{den} \\
e^{i\check{\omega}} &\in \mathcal{G}_\Xi
\end{aligned} \tag{51}$$

γ_0 is the density of the state $|\Psi_0\rangle$, $e^{\check{\mathcal{V}}_{den}}$ produces states of different density, $e^{\check{\mathcal{V}}_{fib}(\gamma)}$ produces states with the same density as $e^{\check{\mathcal{V}}_{den}}|\Psi_0\rangle$ and the shorthand notation for exponentiated integral operators was introduced at eq. (43).

Equations (50), (51) make clear the distinction between the time propagation of the density $\gamma(t)$ and the states. In particular, it is vital that different $\check{\mathcal{V}}_{den}$'s produce states with different densities and that all possible densities γ associated with pure states can be generated by varying $\mathcal{V}_d(\gamma_0)$ such that there is a 1-1 correspondence between $\check{\mathcal{V}}_{den}$ and γ . That is, for each γ there is a unique transformation $e^{\check{\mathcal{V}}_{den}} \in \mathfrak{G}_{den}$ that maps $|\Psi_0\rangle \mapsto |\Psi_\gamma\rangle$, where $|\Psi_\gamma\rangle$ is *some* state that yields γ . Details are in Appendices D and G. Parameterization by $\check{\mathcal{V}}_{den}$ therefore is equivalent to parameterization with respect to γ . Equivalently

$$|\Psi(\mathcal{V}_{fib}(\gamma), \omega, \gamma)\rangle \equiv |\Psi(\mathcal{V}_{fib}(\gamma), \omega, \mathcal{V}_{den})\rangle \quad (52)$$

Notice in this last expression that the roles of $\mathcal{V}_{fib}(\gamma)$ and ω are to determine the time propagation of the states (hence also the density operators) associated with γ . The “fiber factor”

$$e^{\check{\mathcal{V}}_{fib}(\gamma)} e^{i\check{\omega}} \equiv e^{\check{\Upsilon}} \quad (53)$$

selects among elements of the fiber, while $e^{\check{\mathcal{V}}_{den}}$ is the “base factor” that determines which γ will be reached from the initial one. From this perspective, the task of constructing density functionals is equivalent to the task of finding the optimal $\mathcal{V}_{fib}(\gamma)$ and ω for the particular γ . By comparison with the development leading to eq. (25), the equation of motion in terms of this parameterization is

$$\{(\mathcal{V}_{fib}(\gamma), \omega, \gamma), \mathcal{E}\} = (\dot{\mathcal{V}}_{fib}(\gamma), \dot{\omega}, \dot{\gamma}) \quad (54)$$

In Appendix I we show how to utilize the $(\mathcal{V}_{fib}(\gamma), \omega)$ parameterization in a form which emphasizes the complex analytic character of the fiber parameterization via use of the complex fiber parameter

$$\Upsilon = \mathcal{V}_{fib}(\gamma) + i\omega \quad (55)$$

The result of this parameterization is a different EOM, again a pure density functional dynamical expression, of the same form as eq. (31)

$$\dot{\gamma}(t) = \mathcal{N}_1(\gamma(t)) [\nabla_\gamma \mathcal{E}(\gamma(t)) - \mathcal{N}_2(\gamma(t))] \quad (56)$$

but with different functionals and with the positivity constraint built in. Here $\mathcal{N}_i(\gamma(t))$, $1 \leq i \leq 2$, and $\mathcal{E}(\gamma(t))$ are density functionals determined by the solutions of the equations of motion for the parameterizing functions $\mathcal{V}_{fib}(\gamma, t)$ and $\omega(t)$. Appendix I gives their detailed structure.

The $\mathcal{N}_i(\gamma(t))$ in eq. (56) are fiber quantities that provide information on intra-fiber searches and their propagation in time. Thus they act as time-dependent and density-dependent potentials. $\mathcal{E}(\gamma(t))$ also is an implicit function of fiber factors but an explicit function of γ , hence may be decomposed in conventional KS fashion into kinetic, coulombic, and exchange-correlation parts if that proves useful.

VII. SUMMARY AND CONCLUSIONS

We summarize what has been shown. First, to understand the details of constrained search in the most general way, we established that the space of trace class operators $\mathcal{B}_1(\mathcal{H}^N)$ can be sorted into equivalence classes

$$\{[f]_N; \quad f \in L_1(\mathbb{Z})\} \quad (57)$$

This classification endows $\mathcal{B}_1(\mathcal{H}^N)$ with two fiber bundle structures, one affine $\mathfrak{B}_{aff}$, the other vector $\mathfrak{B}_{vect}$. In $\mathfrak{B}_{aff}$, the fibers are the $[f]_N$ and the base of the bundle is $L_1(\mathbb{Z})$, while in $\mathfrak{B}_{vect}$ the fibers are $\ker \Xi$ and the base is $\ker \Xi^c$. The projection map from the fiber to the base in the affine case is the contraction map Ξ defined in eq. (12)

$$\Xi(Y) = Tr\{\Phi^\dagger \Phi Y\} = f; \quad Y \in \mathcal{B}_1(\mathcal{H}^N); \quad f \in L_1(\mathbb{Z}) \quad (58)$$

and the product of field operators is as in eq. (8). In the vector bundle the corresponding projection map is the projection onto the unique element $\Gamma(f)$ of $\ker \Xi^c$ that corresponds to f under the mapping Ξ . The fibers of the affine bundle (the equivalence classes $[f]_N$) correspond to $\{\ker \Xi\} \times \{\Gamma(f)\}$ i.e. the set $\{(k, \Gamma(f)); k \in \ker \Xi, \text{ fixed } \Gamma(f) \in \ker \Xi^c\}$ which is a fiber and base point of the vector bundle $\mathfrak{B}_{vect}$.

Second, because Ξ is a linear map, its kernel is a linear subspace of $\mathcal{B}_1(\mathcal{H}^N)$. Because of the fiber bundle structures, the kernel plays a central role in executing the part of constrained search that corresponds to searching at a specified density. Detailed knowledge of the structure of the kernel, in particular its invariances, therefore provides an entirely new route to construction of functionals.

Third, because physical systems are characterized by density operators, i.e. positive trace class operators $Y \in \mathcal{B}_1(\mathcal{H}^N)_+ \equiv \{Y \in \mathcal{B}_1(\mathcal{H}^N), Y \geq 0\}$, we focused upon fibers corresponding to $\{[\gamma]_N, \gamma \geq 0\}$. Then we demonstrated [Appendix E, eq. (E28)] that, for pure states (and all physical densities correspond to pure states, though not necessarily ground states), any density operator can be decomposed as

$$D^N = D_{K_{OD}}^N + D_{K_D}^N + \Gamma(\gamma) \quad (59)$$

Fourth, each of these three contributions has specific invariance properties related to whether the operators are diagonal or off-diagonal in the coordinate eigenfunction representation and to whether they lie in $\ker \Xi$ or not; see Appendices E and F. For time-dependent DFT, these invariances provide a Lie group structure that can be used to parameterize the dynamics via the TDVP. For both time-dependent and time-independent DFT, the invariances provide a new set of tools for constructing functionals. We distinguish “constructing” from “devising” because Appendices G and I comprise a *systematic* way to execute intra- and inter-fiber searches via Lie groups of transformations. In both time-dependent and time-independent forms of the theory, explicit knowledge of the optimal result of the intra-fiber search is identical with determination of the exact functional for the density which labels that particular fiber. When restricted to a particular form of pure state, such knowledge is equivalent to determination of the proper functional to associate with that type of state.

The analytic form of those Lie group transformations is suggestive of certain well-known cluster expansions.[?] This aspect is taken up in Part II (following paper) as one of several means to obtain workable systematic approximations. Also treated there are the related issues of the various classical approximations for the nuclear motion (clamped nucleus, adiabatic motion on electronic potential, narrow wave-packet, etc.). On that topic we remark here only that an effort to extract one of those results simply as the limiting case of the exact dynamical equations must be treated with care because such a limit requires imposition of a constraint on the Hamiltonian (or action), hence amounts to invoking a different Hamiltonian to derive the dynamical equations.

Finally, the challenge of constructing physically realistic, yet tractable time-dependent functionals is evident from the intricate structure of the $\mathcal{N}_i(\gamma(t))$ in eq. (56). Further, because that structure is related directly to the metric functions η , it is clear that approximate functionals constructed on physical grounds or by generalization from functionals used

successfully in the time-independent case actually must involve some implicit, unanalyzed assumptions about that metric.

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APPENDIX A: PARTITIONED EQUATION OF MOTION IN THE $\mathbf{c}$ PARAMETERIZATION

Eq. (30), when rewritten in a form analogous with eq. (21), has the obvious matrix partitioning

$$\begin{bmatrix} \eta_{\mathbf{c}\mathbf{c}}(t) & \eta_{\mathbf{c}\gamma}(t) \\ \eta_{\gamma\mathbf{c}}(t) & \eta_{\gamma\gamma}(t) \end{bmatrix} \begin{bmatrix} \dot{\mathbf{c}} \\ \dot{\gamma} \end{bmatrix} = \begin{bmatrix} \nabla_{\mathbf{c}}\mathcal{E}(t) \\ \nabla_{\gamma}\mathcal{E}(t) \end{bmatrix} \quad (\text{A1})$$

This is an operator notation and $\nabla_{\lambda}\mathcal{E}(t) = \delta\mathcal{E}/\delta\lambda$. With $\lambda_i = \mathbf{c}$ or γ , the components of the metric matrix η are

$$\eta_{\lambda_1, \lambda_2}(t, \mathbf{y}, \mathbf{y}') = i \left\{ \frac{\delta}{\delta\tilde{\lambda}_1(\mathbf{y})} \frac{\delta}{\delta\lambda_2(\mathbf{y}')} - \frac{\delta}{\delta\lambda_2(\mathbf{y})} \frac{\delta}{\delta\tilde{\lambda}_1(\mathbf{y}')} \right\} \ln S|_{\substack{\tilde{\lambda}_1=\lambda_1 \\ \tilde{\lambda}_2=\lambda_2}} \quad (\text{A2})$$

Here

$$S \equiv \langle \Psi(\tilde{\mathbf{c}}(t), \tilde{\gamma}(t)) | \Psi(\mathbf{c}(t), \gamma(t)) \rangle \quad (\text{A3})$$

and the analogue to the classical Hamiltonian (recall eq. (23)) is

$$\mathcal{E}(t) = \frac{\langle \Psi(\mathbf{c}(t), \gamma(t)) | H | \Psi(\mathbf{c}(t), \gamma(t)) \rangle}{\langle \Psi(\mathbf{c}(t), \gamma(t)) | \Psi(\mathbf{c}(t), \gamma(t)) \rangle} \quad (\text{A4})$$

Coupled equations of motion follow

$$\begin{aligned} \dot{\mathbf{c}}(t) &= \eta_{\mathbf{c}\mathbf{c}}^{-1}(t) [\nabla_{\mathbf{c}}\mathcal{E}(t) - \eta_{\mathbf{c}\gamma}(t)\dot{\gamma}] \\ \dot{\gamma}(t) &= [\eta_{\gamma\gamma}(t) - \eta_{\gamma\mathbf{c}}(t)\eta_{\mathbf{c}\mathbf{c}}^{-1}(t)\eta_{\mathbf{c}\gamma}(t)]^{-1} \\ &\quad \times [\nabla_{\gamma}\mathcal{E}(t) - \eta_{\gamma\mathbf{c}}(t)\eta_{\mathbf{c}\mathbf{c}}^{-1}(t)\nabla_{\mathbf{c}}\mathcal{E}(t)] \end{aligned} \quad (\text{A5})$$

Under the assumption that $\eta_{\mathbf{c}\mathbf{c}}^{-1}(t)$ exists, denote by $\mathbf{c}_{\#}(t)$ the solution of the first of these for the correct density $\gamma_{\#}(t)$. Then the exact equation of motion for the temporal development

of the density is

$$\dot{\gamma}(t) = \tilde{\mathcal{N}}_1(\gamma(t)) \left[\nabla_{\gamma} \mathcal{E}(\gamma(t)) - \tilde{\mathcal{N}}_2(\gamma(t)) \right] \quad (\text{A6})$$

The functionals are defined as

$$\begin{aligned} \tilde{\mathcal{N}}_1(\gamma(t)) &= [\eta_{\gamma\gamma}(\mathbf{c}_{\#}(t), \gamma(t)) - \eta_{\gamma\mathbf{c}}(\mathbf{c}_{\#}(t), \gamma(t)) \eta_{\mathbf{c}\mathbf{c}}^{-1}(\mathbf{c}_{\#}(t), \gamma(t)) \times \\ &\quad \eta_{\mathbf{c}\gamma}(\mathbf{c}_{\#}(t), \gamma(t))]^{-1} \\ \tilde{\mathcal{N}}_2(\gamma(t)) &= \eta_{\gamma\mathbf{c}}(\mathbf{c}_{\#}(t), \gamma(t)) \eta_{\mathbf{c}\mathbf{c}}^{-1}(\mathbf{c}_{\#}(t), \gamma(t)) \nabla_{\mathbf{c}} \mathcal{E}(\mathbf{c}_{\#}(t), \gamma(t)). \end{aligned} \quad (\text{A7})$$

Any singularities are associated with bifurcation points of the time evolution of the system.

APPENDIX B: UNITARY GROUPS AND INVARIANCES OF THE KERNEL

Here we give details as to why it is necessary to use the general linear group rather than the unitary group to describe the invariances of the kernel. The time-dependence is inconsequential to the discussion, so is suppressed.

If consideration were restricted to unitary operators, the invariance group of the kernel would be

$$\mathcal{G}_{\text{ker}}^U = \{U \in \mathcal{U}(\mathcal{H}^N); \text{Tr}\{\Phi^\dagger \Phi U Y U^\dagger\} = \text{Tr}\{\Phi^\dagger \Phi Y\} = 0 \forall Y \in \ker \Xi\} \quad (\text{B1})$$

where $\mathcal{U}(\mathcal{H}^N)$ is the group of unitary operators on N -particle space. The condition expressed in eq. (B1) does not imply that transformations belonging to $\mathcal{G}_{\text{ker}}^U$ send elements of an equivalence class to itself

$$\mathcal{G}_{\text{ker}}^U : [f]_N \rightarrow [f']_N; \quad \text{in general } f \neq f' \quad (\text{B2})$$

On the other hand the condition that defines the unitary group

$$\mathcal{G}_{\Xi} : \ker \Xi \rightarrow \ker \Xi \quad (\text{B3})$$

that does send equivalence classes to equivalence classes is given by

$$\mathcal{G}_{\Xi} = \{U \in \mathcal{U}(\mathcal{H}^N); \text{Tr}\{\Phi^\dagger \Phi U Y U^\dagger\} = \text{Tr}\{\Phi^\dagger \Phi Y\} \quad \forall Y \in \mathcal{B}_1(\mathcal{H}^N)\} \quad (\text{B4})$$

Evidently from eq. (B4)

$$[U(\omega), \Phi^\dagger \Phi] = 0 \quad (\text{B5})$$

and $\mathcal{G}_\Xi$ is obviously a subgroup of $\mathcal{G}_{\text{ker}}^U$ and can be represented by transformations of the form

$$\begin{aligned}\mathcal{G}_\Xi &\subset \mathcal{G}_{\text{ker}}^U = \{U(\omega); \hat{U}(\omega)(Y) \equiv U(\omega)YU^\dagger(\omega); \\ U(\omega) &= e^{i \int \omega(\kappa_1, \dots, \kappa_N) \Phi^\dagger(\kappa_1) \Phi(\kappa_1) \dots \Phi^\dagger(\kappa_N) \Phi(\kappa_N) d\kappa_1 \dots d\kappa_N}, \\ \omega(\kappa_1, \dots, \kappa_N) &\in \mathbb{R}\}\end{aligned}\tag{B6}$$

One can see that $\mathcal{G}_\Xi$ is the largest commutative unitary subgroup of $\mathcal{G}_{\text{ker}}^U$ with these properties. Because of eq. (18), it follows that fibers associated with states are mapped into themselves by $\hat{U}(\omega)$

$$Y \in [\gamma]_N \Leftrightarrow \hat{U}(\omega)Y \in [\gamma]_N \tag{B7}$$

However, given a reference element $Y_{ref} \in [\gamma]_N$ corresponding to a pure state $|\Psi_{ref}\rangle$ in the fiber $[\gamma]_N$, it is *not* the case that all the rest of the pure-state segment of the fiber is generated by application of elements of $\mathcal{G}_\Xi$, (see Appendix F), i.e. this group does *not* act homogeneously on the set of pure states in $[\gamma]_N$. This also shows that *there is no intra fiber unitary subgroup of $\mathcal{G}_{\text{ker}}^U$ that acts homogeneously on the set of pure states contained within a fiber*. The physical insight is that a transformation such as eq. (B6) never can generate the equivalent of a Jastrow or Hylleras factor in the Hilbert space of pure states. If one drops the unitary requirement in eq. (B1) it is possible find transformations, that are diagonal in the coordinate representation (like those of $\mathcal{G}_\Xi$), which produce all pure states with a given density, γ , from a pure reference state $|\Psi\rangle$ of that density. These transformations are of the form

$$T(\mathcal{V}, \omega) = e^{\int \{\mathcal{V}(\kappa_1, \dots, \kappa_N) + i\omega(\kappa_1, \dots, \kappa_N)\} \Phi^\dagger(\kappa_1) \Phi(\kappa_1) \dots \Phi^\dagger(\kappa_N) \Phi(\kappa_N) d\kappa_1 \dots d\kappa_N} \tag{B8}$$

where there are certain restriction placed on $\mathcal{V}(\kappa_1, \dots, \kappa_N)$ (see Appendix F). We denote the group formed by these intra fiber transformations $\mathfrak{G}_{fib}$ in the main body of this article. With a weaker set of restrictions, (see Appendix F), such maps also transform $\text{ker } \Xi \rightarrow \text{ker } \Xi$, and form a subgroup $\mathfrak{G}_{\text{ker}} \subset \mathfrak{G}_\Xi$, and with a third set of restrictions, one forms a group of inter fiber transformations $\mathfrak{G}_{den}$ (see Appendix F).

APPENDIX C: POSITIVE MAPS OF TRACE CLASS OPERATORS

Recall that a trace class operator $Y \in \mathcal{B}_1(\mathcal{H})$, with $\mathcal{H}$ any Hilbert space, is positive $Y \geq 0$ if

$$\langle v|Yv \rangle \geq 0 \quad \forall v \in \mathcal{H} \quad (\text{C1})$$

which is true if and only if all the eigenvalues of $Y \geq 0$. In turn this requirement is true if and only if

$$Y = QQ^\dagger, Q \in \mathcal{B}_2(\mathcal{H}) \quad (\text{C2})$$

and $\mathcal{B}_2(\mathcal{H})$ is the space of Hilbert-Schmidt operators.

A linear map of trace class operators $\hat{T} : \mathcal{B}_1(\mathcal{H}) \mapsto \mathcal{B}_1(\mathcal{H})$ is positive if $\hat{T}(Y) \geq 0$ for $Y \geq 0$. In particular, $\hat{T}(Y) = TYT^\dagger$ is a positive map $\mathcal{B}_1(\mathcal{H}) \mapsto \mathcal{B}_1(\mathcal{H})$ for any bounded operator T , i.e., for such a T

$$\hat{T}(Y) \in \mathcal{B}_1(\mathcal{H})_+ \quad \text{if } Y \in \mathcal{B}_1(\mathcal{H})_+ \quad (\text{C3})$$

where $\mathcal{B}_1(\mathcal{H})_+$ is the cone of positive trace class operators [?]. Cones facilitate analysis because they define order relationships. That is, $A > B$ if $A - B \in \mathcal{B}_1(\mathcal{H})_+$. The set of positive operators is a cone, not just a convex set, as normalization does not have to be constant.

If T is an invertible bounded operator, i.e. $T \in GL(\mathcal{H})$, then

$$T = e^A e^{iB} \quad (\text{C4})$$

with $A^\dagger = A$ and $B^\dagger = B$. Clearly e^{iB} is unitary while e^A is invertible and positive. The set $\{e^{iB}\}$ forms the unitary subgroup $\mathcal{U}(\mathcal{H})$ of $GL(\mathcal{H})$, while the set $\{e^A\}$ comprises the left coset $GL(\mathcal{H})/\mathcal{U}(\mathcal{H})$ of $\mathcal{U}(\mathcal{H})$ in $GL(\mathcal{H})$. Denote the maps $\hat{T}$ generated by $T \in GL(\mathcal{H})$ as $\widehat{GL(\mathcal{H})}$ and observe that they map $\mathcal{B}_1(\mathcal{H}) \mapsto \mathcal{B}_1(\mathcal{H})$ and $\mathcal{B}_2(\mathcal{H}) \mapsto \mathcal{B}_2(\mathcal{H})$. These are used in the next Appendix.

APPENDIX D: GROUP ORBITS OF DENSITY OPERATORS

Once again the time-dependence appears only as a parameterization that is inconsequential to the discussion, so is omitted. The transformation

$$\begin{aligned}\hat{T} : \mathcal{B}_1(\mathcal{H}) &\mapsto \mathcal{B}_1(\mathcal{H}) \\ \hat{T}(Y) &= TYT^\dagger\end{aligned}\tag{D1}$$

maintains the rank of the operator Y , that is $\hat{T}(Y)$ has the same number of non-zero eigenvalues as Y itself. Particularizing to density operators D , their cone $\mathcal{B}_1(\mathcal{H})_+$ can be classified by the rank of $D \in \mathcal{B}_1(\mathcal{H})_+$. Operators for pure states have unit rank and correspond to vectors in $\mathcal{H}$

$$\text{rank}(D) = 1 \Leftrightarrow D = \frac{|\Psi\rangle\langle\Psi|}{\langle\Psi|\Psi\rangle}; |\Psi\rangle \in \mathcal{H}\tag{D2}$$

All rank 1 density operators lie in the same orbit of $\widehat{GL(\mathcal{H})}$ (see Appendix C for definition) in $\mathcal{B}_1(\mathcal{H})$. That is, if D_0 is an arbitrary, specified pure state density operator, then all other pure state density operators D are generated by

$$D = \hat{T}(D_0); \quad \hat{T} \in \widehat{GL(\mathcal{H})}; \quad \text{rank}(D_0) = 1\tag{D3}$$

Ensemble states correspond to density operators with rank greater than unity. As in the pure state case, ensemble density operators of a given rank all lie on one orbit of $\widehat{GL(\mathcal{H})}$. Thus, if $D_{p,0}$ is an arbitrary, specified ensemble state density operator of rank p , then all other ensemble state density operators of the same rank D_p are generated by

$$D_p = \hat{T}(D_{p,0}); \quad \hat{T} \in \widehat{GL(\mathcal{H})}; \quad \text{rank}(D_{p,0}) = p\tag{D4}$$

APPENDIX E: ANALYTICAL STRUCTURE OF THE KERNEL

This Appendix treats characteristics of $Y \in \ker \Xi$. As before, the time-dependence appears only as a parameterization that is inconsequential to the discussion, so is omitted. It is useful for present purposes to begin by expressing all operators belonging to $\mathcal{B}_1(\mathcal{H}^N)$ (recall $N = N_e + N_n$) in terms of the ordered generalized operator basis [?]]

$$\left\{ |\mathbf{x}_1 \dots \mathbf{x}_{N_e}\rangle_{ord} \langle \mathbf{x}'_1 \dots \mathbf{x}'_{N_e}|_{ord} \otimes |\mathbf{X}_1 \dots \mathbf{X}_{N_n}\rangle_{ord} \langle \mathbf{X}'_1 \dots \mathbf{X}'_{N_n}|_{ord} \right\}\tag{E1}$$

with

$$\mathbf{x}, \mathbf{x}', \mathbf{X}, \mathbf{X}' \in \mathbb{F} \quad (\text{E2})$$

where $\mathbb{F} = \mathbb{R}^3 \otimes \mathbb{C}^2$ if the space-spin analysis is pursued and $\mathbb{F} = \mathbb{R}^3$ if spin is suppressed (whence the states are coordinate eigenfunctions). The subscript “*ord*” denotes the ordered basis

$$\mathbf{x}_1 \leq \mathbf{x}_2 \leq \mathbf{x}_3 \leq \dots \quad (\text{E3})$$

with the understanding that strict inequality holds only for each species of fermions. Shortly a shift to the unordered basis will be advantageous. For either basis, a compressed notation, analogous with eq. (40), is convenient

$$\begin{aligned} & |\mathbf{x}_1 \dots \mathbf{x}_{N_e}\rangle \langle \mathbf{x}'_1 \dots \mathbf{x}'_{N_e}| \otimes |\mathbf{X}_1 \dots \mathbf{X}_{N_n}\rangle \langle \mathbf{X}'_1 \dots \mathbf{X}'_{N_n}| \\ & \equiv |\kappa_1 \dots \kappa_{N_e}, \kappa_{N_e+1} \dots \kappa_N\rangle \langle \kappa'_1 \dots \kappa'_{N_e}, \kappa'_{N_e+1} \dots \kappa'_N| \\ & \equiv |\mathbf{x}, \mathbf{X}\rangle \langle \mathbf{x}', \mathbf{X}'|, \end{aligned} \quad (\text{E4})$$

The last form (recall eq. (3)) will be used only when there is no ambiguity.

Any operator $Y \in \mathcal{B}_1(\mathcal{H}^N)$ can be expanded in terms of the ordered basis as

$$\begin{aligned} Y &= \int d\kappa_1, \dots, d\kappa_N, d\kappa'_1, \dots, d\kappa'_N \tilde{Y}(\kappa_1, \dots, \kappa_N; \kappa'_1, \dots, \kappa'_N) \\ &\quad \times |\kappa_1 \dots \kappa_N\rangle_{ord} \langle \kappa'_1 \dots \kappa'_N|_{ord} \end{aligned} \quad (\text{E5})$$

The operator integral kernel with respect to the ordered basis can be expanded [?] as

$$\begin{aligned} \tilde{Y}(\kappa_1, \dots, \kappa_N, \kappa'_1, \dots, \kappa'_N) &= \sum_{i=1}^{\infty} \alpha_i \bar{\varphi}_i(\kappa_1, \dots, \kappa_N) \psi_i(\kappa'_1, \dots, \kappa'_N); \\ \alpha_i &\geq 0; \quad \sum_{i=1}^{\infty} \alpha_i < \infty \end{aligned} \quad (\text{E6})$$

where the basis functions $\{\varphi_i, \psi_i\}$ are specific to the operator Y in the sense of spanning its range and domain and satisfying

$$\begin{aligned} \int d\kappa_1, \dots, d\kappa_N, \bar{\varphi}_i(\kappa_1, \dots, \kappa_N) \varphi_j(\kappa_1, \dots, \kappa_N) &= \delta_{ij} \\ \int d\kappa_1, \dots, d\kappa_N, \bar{\psi}_i(\kappa_1, \dots, \kappa_N) \psi_j(\kappa_1, \dots, \kappa_N) &= \delta_{ij} \end{aligned} \quad (\text{E7})$$

It is easier to consider the contraction map, the central quantity, in the *unordered* basis. In it, the operator integral kernel is

$$Y(\kappa_1, \dots, \kappa_N, \kappa'_1, \dots, \kappa'_N) = \frac{(-1)^{(P_e+P_n)}}{(N_e!N_n!)^2} \mathcal{P} \tilde{Y}(\kappa_1, \dots, \kappa_N, \kappa'_1, \dots, \kappa'_N) \quad (\text{E8})$$

where $\mathcal{P}^{-1}$ is the permutation operator that generates the ordered strings from the unordered ones

$$\begin{aligned}\mathcal{P}^{-1}\{\kappa_1, \dots, \kappa_N\} &= \{\kappa_1, \dots, \kappa_N\}_{ord} \\ \mathcal{P}^{-1}\{\kappa'_1, \dots, \kappa'_N\} &= \{\kappa'_1, \dots, \kappa'_N\}_{ord}\end{aligned}\quad (\text{E9})$$

P_e is the number of permutations of the electron coordinates, and P_n is either the number of permutations of the nuclear coordinates if the nuclei are fermions or 0 if they are bosons.

In the unordered basis the contraction map Ξ eq. (12) on any operator Y is

$$\begin{aligned}& Tr\{\Phi^\dagger(\mathbf{x})\Phi(\mathbf{x})\Phi^\dagger(\mathbf{X})\Phi(\mathbf{X})Y\} \\ &= \int d\kappa_1, \dots, d\kappa_N d\kappa'_1, \dots, d\kappa'_N d\kappa''_1, \dots, d\kappa''_N \\ &\quad \times \langle \kappa''_1, \dots, \kappa''_N | \Phi^\dagger(\mathbf{x})\Phi(\mathbf{x})\Phi^\dagger(\mathbf{X})\Phi(\mathbf{X}) | \kappa_1, \dots, \kappa_N \rangle \\ &\quad \times \langle \kappa'_1, \dots, \kappa'_N | \kappa''_1, \dots, \kappa''_N \rangle Y(\kappa_1, \dots, \kappa_N; \kappa'_1, \dots, \kappa'_N) \\ &= N_e N_n \int d\mathbf{x}'_2 \dots, d\mathbf{x}'_{N_e} d\mathbf{X}'_2, \dots, d\mathbf{X}'_{N_n} \\ &\quad \times Y(\mathbf{x}, \mathbf{x}_2 \dots, \mathbf{x}_{N_e} \mathbf{X}, \mathbf{X}_2 \dots, \mathbf{X}_{N_n}; \mathbf{x}, \mathbf{x}'_2 \dots, \mathbf{x}'_{N_e} \mathbf{X}, \mathbf{X}'_2 \dots, \mathbf{X}'_{N_n})\end{aligned}\quad (\text{E10})$$

The elements of $\ker \Xi$ become evident by writing

$$\begin{aligned}Y(\kappa_1, \dots, \kappa_N; \kappa'_1, \dots, \kappa'_N) &= Y_d(\kappa_1, \dots, \kappa_N) \prod_{j=1}^N \delta(\kappa_j - \kappa'_j) \\ &\quad + Y_{OD}(\kappa_1, \dots, \kappa_N; \kappa'_1, \dots, \kappa'_N)\end{aligned}\quad (\text{E11})$$

$$\begin{aligned}Y_{OD}(\kappa_1, \dots, \kappa_N; \kappa_1, \dots, \kappa_N) &= 0 \\ Y_d(\kappa_1, \dots, \kappa_N) &\equiv Y(\kappa_1, \dots, \kappa_N; \kappa_1, \dots, \kappa_N)\end{aligned}\quad (\text{E12})$$

and the caveat that the product of delta functions in eq. (E11) implicitly includes all the permutations of coordinates (because of working in the unordered basis). Note that Y_d is symmetric with respect to interchange among the electron coordinates themselves and similarly for the nuclear coordinates. Substitution of (E11) in (E10) gives

$$\begin{aligned}& Tr\{\Phi^\dagger(\mathbf{x})\Phi(\mathbf{x})\Phi^\dagger(\mathbf{X})\Phi(\mathbf{X})Y\} = \\ & N_e N_n \int d\mathbf{x}'_2 \dots, d\mathbf{x}'_{N_e} d\mathbf{X}'_2, \dots, d\mathbf{X}'_{N_n} Y_d(\mathbf{x}, \mathbf{x}_2 \dots, \mathbf{x}_{N_e} \mathbf{X}, \mathbf{X}_2 \dots, \mathbf{X}_{N_n})\end{aligned}\quad (\text{E13})$$

An alternative way to state this is that the contraction map sends all off-diagonal basis operators to zero

$$\begin{aligned} \Xi \{ |\kappa_1 \dots \kappa_N\rangle \langle \kappa'_1 \dots \kappa'_N| \} &= 0 \\ \text{for } \{ \kappa_1 \dots \kappa_{N_e} \} &\neq \{ \kappa'_1 \dots \kappa'_{N_e} \} \text{ or } \{ \kappa_1 \dots \kappa_{N_n} \} \neq \{ \kappa'_1 \dots \kappa'_{N_n} \} \end{aligned} \quad (\text{E14})$$

Three cases are obvious immediately from eq. (E13); two belong to $\ker \Xi$, one does not. First, there are pure off-diagonal operators

$$Y = Y_{OD} ; (Y_d = 0) \Rightarrow Y \in \ker \Xi \quad (\text{E15})$$

Then there are diagonal operators with a special property

$$\begin{aligned} Y &= Y_d; (Y_{OD} = 0) \\ \int d\mathbf{x}_2 \dots, d\mathbf{x}_{N_e} d\mathbf{X}_2, \dots, d\mathbf{X}_{N_n} Y_d(\mathbf{x}, \mathbf{x}_2 \dots, \mathbf{x}_{N_e} \mathbf{X}, \mathbf{X}_2 \dots, \mathbf{X}_{N_n}) &= 0 \\ \Rightarrow Y &\in \ker \Xi \end{aligned} \quad (\text{E16})$$

Finally, there are diagonal operators that do not belong to $\ker \Xi$

$$\begin{aligned} Y &= Y_d; (Y_{OD} = 0) \\ 0 &\neq \int d\mathbf{x}_2 \dots, d\mathbf{x}_{N_e} d\mathbf{X}_2, \dots, d\mathbf{X}_{N_n} Y_d(\mathbf{x}, \mathbf{x}_2 \dots, \mathbf{x}_{N_e} \mathbf{X}, \mathbf{X}_2 \dots, \mathbf{X}_{N_n}) \\ \Rightarrow Y &\notin \ker \Xi \end{aligned} \quad (\text{E17})$$

The space K_{OD} generated by operators satisfying (E15) is a subspace of $\ker\{\Xi\}$

$$\begin{aligned} K_{OD} &\subset \ker\{\Xi\} \\ K_{OD} &= \text{linear span } \{Y = Y_{OD}, Y_d = 0\} \end{aligned} \quad (\text{E18})$$

The remaining part K_D of $\ker\{\Xi\}$ is comprised of diagonal operators that satisfy eq. (E16)

$$\begin{aligned} K_D &\subset \ker\{\Xi\}; \text{ where } K_D = \text{linear span } \{Y = Y_d, Y_{OD} = 0; \\ 0 &= \int d\mathbf{x}_2, \dots, d\mathbf{x}_{N_e} d\mathbf{X}_2, \dots, d\mathbf{X}_{N_n} Y_d(\mathbf{x}, \mathbf{x}_2, \dots, \mathbf{x}_{N_e} \mathbf{X}, \mathbf{X}_2, \dots, \mathbf{X}_{N_n}) \} \end{aligned} \quad (\text{E19})$$

In detail, for K_D , Y_d obeys at least one of the two conditions

$$\begin{aligned} 0 &= \int d\mathbf{x}_{i_1}, \dots, d\mathbf{x}_{i_{N_e-1}}, Y_d(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n}) \\ 0 &= \int d\mathbf{X}_{j_1}, \dots, d\mathbf{X}_{j_{N_n-1}}, Y_d(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n}) \end{aligned} \quad (\text{E20})$$

with

$$\begin{aligned} \{i_1 \dots i_{N_e-1}\} &\subset \{1 \dots N_e\} \\ \{j_1 \dots j_{N_n-1}\} &\subset \{1 \dots N_n\} \end{aligned} \quad (\text{E21})$$

Thus the kernel of Ξ can be expressed as

$$\ker\{\Xi\} = K_{OD} \oplus K_D \quad (\text{E22})$$

with the two components orthogonal with respect to the Hilbert-Schmidt inner product

$$\text{Tr}\{k_{OD}^\dagger k_D\} = 0 \text{ or } k_{OD} \in K_{OD}, k_D \in K_D \quad (\text{E23})$$

This result follows from

$$\begin{aligned} \text{Tr}\{|\kappa_1 \dots \kappa_N\rangle \langle \kappa_1 \dots \kappa_N| |\kappa'_1 \dots \kappa'_N\rangle \langle \kappa''_1 \dots \kappa''_N|\} &= \\ \langle \kappa_1 \dots \kappa_N | \kappa'_1 \dots \kappa'_N \rangle \langle \kappa''_1 \dots \kappa''_N | \kappa_1 \dots \kappa_N \rangle &= 0 \end{aligned} \quad (\text{E24})$$

which holds when both

$$\{\kappa'_1 \dots \kappa'_{N_e}\} \neq \{\kappa''_1 \dots \kappa''_{N_e}\} \quad (\text{E25})$$

and

$$\{\kappa'_{N_e+1} \dots \kappa'_N\} \neq \{\kappa''_{N_e+1} \dots \kappa''_N\} \quad (\text{E26})$$

In either case the primed and double primed sets of coordinates have at least one difference, hence both cannot be equal to $\{\kappa_1 \dots \kappa_N\}$.

Eq. (E17) makes clear that the only operators $Y \in \mathcal{B}_1(\mathcal{H}^N)$ that do not lie in $\ker \Xi$ are diagonal operators that do not belong to K_D . These form the subspace (of $\mathcal{B}_1(\mathcal{H}^N)$)

$$\begin{aligned} S_D &= \text{linear span}\{Y = Y_d, Y_{OD} = 0; \\ 0 &\neq \int d\mathbf{x}_2 \dots, d\mathbf{x}_{N_e} d\mathbf{X}_2, \dots, d\mathbf{X}_{N_n} Y_d(\mathbf{x}, \mathbf{x}_2 \dots, \mathbf{x}_{N_e} \mathbf{X}, \mathbf{X}_2 \dots, \mathbf{X}_{N_n})\} \end{aligned} \quad (\text{E27})$$

Therefore the space of trace class operators has the direct sum decomposition

$$\mathcal{B}_1(\mathcal{H}^N) = K_{OD} \oplus K_D \oplus S_D \quad (\text{E28})$$

APPENDIX F: VECTOR BUNDLE TRANSFORMATIONS

This Appendix develops pertinent properties of transformations $TYT^\dagger$ of trace class operators Y in $K_{OD} \oplus K_D \oplus S_D$ by elements T of the General Linear Group $GL(\mathcal{H}^N)$ of $\mathcal{H}^N$. In the coordinate eigenfunction basis, the transformation $TYT^\dagger$ reads

$$TYT^\dagger \equiv \hat{T}Y = \int dz dz' [TYT^\dagger(z, z')] |z\rangle\langle z'| \quad (F1)$$

where the transformed integral kernel is

$$TYT^\dagger(z, z') = \int dz'' dz''' T(z, z'') Y(z'', z''') T^*(z''', z') \quad (F2)$$

Clearly the only transforms that can preserve the structure described by eqs. (E18), (E19), (E27) have diagonal representations

$$\begin{aligned} T(z, z') &= \delta(z - z') T(z) \\ T(z) &= T(\mathbf{x}, \mathbf{X}) \end{aligned} \quad (F3)$$

with $T(\mathbf{z})$ a species symmetric function with respect to interchange among electronic coordinates and similarly for nuclear coordinates (within each species).

The explicit form of the diagonal operator is

$$\begin{aligned} T &= e^{\int \mathcal{V}(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n}) \prod_{j=1}^{N_e} \Phi_e^\dagger(x_j) \Phi_e(x_j) \prod_{k=1}^{N_n} \Phi_n^\dagger(X_k) \Phi_n(X_k) \prod_{l=1}^{N_e} dx_l \prod_{m=1}^{N_n} dX_m} \\ &\times e^{i \int \omega(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n}) \prod_{q=1}^{N_e} \Phi_e^\dagger(x_q) \Phi_e(x_q) \prod_{r=1}^{N_n} \Phi_n^\dagger(X_r) \Phi_n(X_r) \prod_{s=1}^{N_e} dx_s \prod_{t=1}^{N_n} dX_t} \end{aligned} \quad (F4)$$

When T operates upon an element of the basis of many-body coordinate eigenfunctions, the result is

$$\begin{aligned} T |y_1 \dots y_{N_e} Y_1 \dots Y_{N_n}\rangle \\ = e^{\mathcal{V}(y_1 \dots y_{N_e} Y_1 \dots Y_{N_n})} e^{i\omega(y_1 \dots y_{N_e} Y_1 \dots Y_{N_n})} |y_1 \dots y_{N_e} Y_1 \dots Y_{N_n}\rangle \end{aligned} \quad (F5)$$

because

$$\begin{aligned} \Phi_e^\dagger(x_j) \Phi_e(x_j) |y_1 \dots y_{N_e} Y_1 \dots Y_{N_n}\rangle &= |y_1 \dots y_{N_e} Y_1 \dots Y_{N_n}\rangle \\ &\text{for } x_j \in \{y_1 \dots y_{N_e}\} \\ &= 0 \text{ otherwise} \end{aligned} \quad (F6)$$

and the corresponding relationship for $\Phi_n^\dagger(X_k)\Phi_n(X_k)$.

For $Y = Y_{K_{OD}} \in K_{OD}$, diagonal transformations $\hat{T}$ map $K_{OD} \mapsto K_{OD}$ with the explicit form of the transformed integral kernel

$$TY_{K_{OD}}T^\dagger(\mathbf{z}, \mathbf{z}') = T(\mathbf{z})Y_{K_{OD}}(\mathbf{z}, \mathbf{z}')T^*(\mathbf{z}') \in K_{OD} \quad (\text{F7})$$

Diagonal transformations that map $K_D \mapsto K_D$ provided that the transformed integral kernel satisfies eq. (E16), which is in fact the defining property for maps that maintain the vector bundle structure of $\mathfrak{B}_{vec}$. This can be written as

$$\begin{aligned} \hat{T} : Y_{K_D} &\mapsto TY_{K_D}T^\dagger; \quad Y_{K_D} \in K_D \\ TY_{K_D}T^\dagger(\mathbf{z}) &= |T(\mathbf{z})|^2 Y_{K_D}(\mathbf{z}) \end{aligned} \quad (\text{F8})$$

provided that

$$|T(\mathbf{z})|^2 = 1 + \phi(\mathbf{z}) \quad (\text{F9})$$

and the species symmetric functions ϕ have the property

$$\begin{aligned} 0 &= \int d\mathbf{x}_{i_1}, \dots, d\mathbf{x}_{i_{N_e-1}}, d\mathbf{X}_{j_1}, \dots, d\mathbf{X}_{j_{N_n-1}} \\ &\times \phi(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n}) Y_d(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n}) \\ &\quad \forall Y_{K_D} \in K_D \end{aligned} \quad (\text{F10})$$

and the indices on the variables of integration obey eq. (E21). These conditions follow directly from eqs. (E16). and (F10) and expresses the property of strong orthogonality, denoted by $\check{\phi} \perp_S K_D$, or equivalently,

$$\check{\phi} \perp_S \forall Y_{K_D} \in K_D \quad (\text{F11})$$

where $\check{\phi}$ is the operator defined by the integral kernel ϕ . In particular, eq. (F10) implies that the pointwise product function $\phi Y_d \in L_1(\mathbf{Z})$, which in turn implies that $\phi \in L_\infty(\mathbf{Z})$ since $Y(\mathbf{z}) \in L_1(\mathbf{Z})$. Here

$$\mathbf{Z} = \mathbb{R}^{3N_e} \times \mathbb{R}^{3N_n} \times \mathbb{C}^{2N_e} \times \mathbb{C}^{2N_n} \quad (\text{F12})$$

Vector bundle transformations built from ϕ satisfy

$$\begin{aligned} \phi &\in L_\infty(\mathbf{Z}) \\ \phi &\perp_S K_D \end{aligned} \quad (\text{F13})$$

and have the property of mapping $K_D \mapsto K_D$, $S_D \mapsto K_D \oplus S_D$ and form the group $\mathcal{G}_{\text{ker}}$. This group together with the unitary group $\mathcal{G}_{\Xi}$ forms the group $\mathfrak{G}_{\text{ker}} = \mathcal{G}_{\text{ker}} \times \mathcal{G}_{\Xi}$, which is a subgroup of the commutative subgroup $\mathcal{C}(\mathcal{H}^N) \subset GL(\mathcal{H}^N)$. The elements of $\mathcal{C}(\mathcal{H}^N)$ are given by

$$T = e^{\{\check{\mathcal{V}} + i\check{\omega}\}} = T_d e^{i\check{\omega}} \quad (\text{F14})$$

where the operator T_d is defined by the integral kernel

$$T_d(\kappa_1 \dots \kappa_N) = \exp[\mathcal{V}(\kappa_1 \dots \kappa_N)] = (1 + \phi(\kappa_1 \dots \kappa_N))^{\frac{1}{2}} \quad (\text{F15})$$

with no restrictions on ϕ . The integral kernel of $\check{\mathcal{V}}$ is given by

$$\mathcal{V}(\kappa_1, \dots, \kappa_N) \equiv \frac{1}{2} \ln(1 + \phi(\kappa_1, \dots, \kappa_N)) \quad (\text{F16})$$

The transformations $\hat{T} : K_{OD} \mapsto K_{OD}$ are determined by elements of $\mathfrak{G}_{\text{ker}}$ and act according to eq. (F7).

We now develop the conditions ϕ must satisfy to determine the subgroups $\mathfrak{G}_{\text{den}}$ and $\mathfrak{G}_{\text{fib}}$ of inter-fiber and intra-fiber maps respectively contained in $\mathfrak{G}_{\text{ker}}$. Once this is accomplished we can construct a non redundant parameterizations of pure states in the equivalence classes $[\gamma]_N$ which can be utilized in the constrained search procedure.

First consider transformations, $\hat{T}_{\text{den}}(\phi)$, between fibers that produce the subgroup $\mathfrak{G}_{\text{den}}$, eqs. (F9), (F13) give the conditions for $\phi(\kappa_1, \dots, \kappa_N)$ to map $S_D \mapsto S_D$. In that case

$$\begin{aligned} & \int \phi(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n}) Y_d(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n}) \\ & \quad \times d\mathbf{x}_2, \dots, d\mathbf{x}_{N_e}, d\mathbf{X}_2, \dots, d\mathbf{X}_{N_n} \\ &= \int \phi(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n}) Y_{S_D}(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n}) \\ & \quad \times d\mathbf{x}_2, \dots, d\mathbf{x}_{N_e}, d\mathbf{X}_2, \dots, d\mathbf{X}_{N_n} \neq 0 \end{aligned} \quad (\text{F17})$$

with

$$Y_d = Y_{K_D} + Y_{S_D} \quad (\text{F18})$$

This corresponds to

$$\Phi_{S_D}(\mathbf{z}_1, \mathbf{z}_2) = \int P_{S_D}(\mathbf{z}_1, \mathbf{z}'_1) \phi(\mathbf{z}'_1) P_{S_D}(\mathbf{z}'_1, \mathbf{z}_2) d\mathbf{z}'_1 \neq 0 \quad \forall \mathbf{z}_1, \mathbf{z}_2 \quad (\text{F19})$$

and hence ϕ is not strongly orthogonal to Y_d , where P_{S_D} is the integral kernel of the projector onto the subspace S_D . In order to obtain the defining representation of $\mathfrak{G}_{den}$ on $K_D \oplus S_D$, the functions ϕ should have zero effect on K_D thus need to satisfy

$$\Phi_{K_D}(z_1, z_2) = \int P_{K_D}(z_1, z'_1) \phi(z'_1) P_{K_D}(z'_1, z_2) dz'_1 = 0 \quad \forall z_1, z_2 \quad (\text{F20})$$

and

$$\Phi_{K_D S_D}(z_1, z_2) = \int P_{K_D}(z_1, z'_1) \phi(z'_1) P_{S_D}(z'_1, z_2) dz'_1 = 0 \quad \forall z_1, z_2 \quad (\text{F21})$$

The effect of these transformations on the off diagonal part of $\mathcal{B}_1(\mathcal{H}^N)$, are given by elements of $\mathcal{G}_\Xi$ that act according to eq. (F7) and is given by

$$\hat{T}_{den}(\phi) Y_{K_{OD}} = e^{\check{V}_{den}(\phi)} Y_{K_{OD}} e^{\check{V}_{den}(\phi)} \quad (\text{F22})$$

In order to produce intra-fiber transformations, $\hat{T}_{fib}(\phi) : K_D \mapsto K_D$, that form the subgroup $\mathcal{G}_{fib}$ the functions ϕ must satisfy the condition

$$\Phi_{S_D}(z_1, z_2) = \int P_{S_D}(z_1, z'_1) \phi(z'_1) P_{S_D}(z'_1, z_2) dz'_1 = 0 \quad \forall z_1, z_2 \quad (\text{F23})$$

The defining representation of $\mathcal{G}_{fib}$ on $K_D \oplus S_D$, is obtained when the functions further satisfy

$$\Phi_{K_D S_D}(z_1, z_2) = \int P_{K_D}(z_1, z'_1) \phi(z'_1) P_{S_D}(z'_1, z_2) dz'_1 = 0 \quad \forall z_1, z_2 \quad (\text{F24})$$

The effect of these transformations on the off diagonal subspace K_{OD} and is given by

$$\hat{T}_{fib}(\phi) Y_{K_{OD}} = e^{\check{V}_{fib}(\phi)} Y_{K_{OD}} e^{\check{V}_{fib}(\phi)} \quad (\text{F25})$$

The group of intra fiber transformations that acts homogeneously on the set of pure states within a fiber is $\mathfrak{G}_{fib} = \mathcal{G}_{fib} \times \mathcal{G}_\Xi$ i.e is constructed from transformations of the form $e^{\check{V}_{fib}(\phi)} e^{i\check{\omega}}$

APPENDIX G: NON REDUNDANT PARAMETERIZATION OF PURE STATES ADAPTED TO $\mathfrak{B}_{vec}$

In order to produce non redundant parameterizations of pure states in $\mathcal{B}_1(\mathcal{H}^N)$ by the groups $\mathfrak{G}_{fib}$ and $\mathfrak{G}_{den}$ one needs to introduce biorthogonal bases for K_{OD} , K_D and S_D . (The space of trace class operators $\mathcal{B}_1(\mathcal{H}^N)$ is not a Hilbert space thus is not self dual, its dual

space is the space of bounded operators $\mathcal{B}(\mathcal{H}^N)$. The basis for K_{OD} , K_D and S_D are

$$K_{OD} = \text{linear span} \left\{ \check{h}_l : \check{h}_l = \int h_l(\mathbf{z}, \mathbf{z}') |\mathbf{z}\rangle \langle \mathbf{z}'| d\mathbf{z} d\mathbf{z}'; \mathbf{z} \neq \mathbf{z}' \right\} \quad (\text{G1})$$

$$K_D = \text{linear span} \left\{ \check{k}_l : \check{k}_l = \int k_l(\mathbf{z}) |\mathbf{z}\rangle \langle \mathbf{z}| d\mathbf{z} \right. \\ \left. \int k_l(\kappa_2, \dots, \kappa_{N_e}, \kappa_{N_e+1}, \dots, \kappa_N) d\kappa_2 \dots d\kappa_{N_e} d\kappa_{N_e+2} \dots d\kappa_N = \mathbf{0} \text{ a.e.} \right\} \quad (\text{G2})$$

$$S_D = \text{linear span} \left\{ \check{s}_l : \check{s}_l = \int s_l(\mathbf{z}) |\mathbf{z}\rangle \langle \mathbf{z}| d\mathbf{z}; \right. \\ \left. \int s_l(\kappa_2, \dots, \kappa_{N_e}, \kappa_{N_e+1}, \dots, \kappa_N) d\kappa_2 \dots d\kappa_{N_e} d\kappa_{N_e+2} \dots d\kappa_N \neq \mathbf{0} \text{ a.e.} \right\} \quad (\text{G3})$$

where $\mathbf{a.e.} \equiv$ "almost everywhere" i.e. except on sets of measure zero. These bases are biorthogonal to the bases $\left\{ \check{k}_\kappa^d, \check{s}_\kappa^d; 1 \leq \kappa \leq \infty \right\}$ of $\mathcal{B}(\mathcal{H}^N)$ i.e

$$\begin{aligned} \left(\check{h}_\kappa^d | \check{h}_l \right) &= \left(\check{k}_\kappa^d | \check{k}_l \right) = \left(\check{s}_\kappa^d | \check{s}_l \right) = \delta_{\kappa l} \\ \left(\check{k}_\kappa^d | \check{s}_l \right) &= \left(\check{s}_\kappa^d | \check{k}_l \right) = \left(\check{h}_\kappa^d | \check{s}_l \right) \\ &= \left(\check{h}_\kappa^d | \check{k}_l \right) = \left(\check{k}_\kappa^d | \check{h}_l \right) = \left(\check{s}_\kappa^d | \check{h}_l \right) = 0 \end{aligned} \quad (\text{G4})$$

and we have introduced the super operator scalar product $(A|B) = \text{Tr} \{ A^\dagger B \}$.

If we are given a reference pure state D_{ref} that has a density γ then it can be decomposed as

$$D_{ref}^N K_{OD} + D_{ref}^N K_D + \Gamma(\gamma) \quad (\text{G5})$$

and we can define

$$\begin{aligned} \check{h}_1 &= D_{ref}^N K_{OD} \\ \check{k}_1 &= D_{ref}^N K_D \\ \check{s}_1 &= \Gamma(\gamma) \end{aligned} \quad (\text{G6})$$

Then transformations that parameterize all other pure states in the fiber $[\gamma]_N$ in terms of this reference and $\mathfrak{G}_{fib}$ in a non redundant fashion can be produced from functions ϕ_{fib} defined as

$$\begin{aligned} \phi_{fib}(\mathbf{z}) &= \sum_{1 \leq \kappa \leq \infty} \phi_{fib\kappa} \left(|\mathbf{z}\rangle \langle \mathbf{z}| | \check{k}_\kappa \right) (D_{ref}^{Nd} K_D | \mathbf{z}\rangle \langle \mathbf{z}|) \\ \phi_{fib\kappa} &= \int k_\kappa^{d*}(\mathbf{z}) \phi_{fib}(\mathbf{z}) D_{ref}^N K_D(\mathbf{z}) d\mathbf{z}; \quad 1 \leq \kappa \leq \infty \end{aligned} \quad (\text{G7})$$

where they also satisfy

$$\begin{aligned}
\int s_\iota^{d*}(\mathbf{z}) \phi_{fib}(\mathbf{z}) s_\kappa(\mathbf{z}) d\mathbf{z} &= 0 \quad \forall \iota, \kappa \\
\int k_\iota^{d*}(\mathbf{z}) \phi_{fib}(\mathbf{z}) s_\kappa(\mathbf{z}) d\mathbf{z} &= 0 \quad \forall \iota, \kappa \\
\int s_\iota^{d*}(\mathbf{z}) \phi_{fib}(\mathbf{z}) k_\kappa(\mathbf{z}) d\mathbf{z} &= 0 \quad \forall \iota, \kappa \\
\int k_\iota^{d*}(\mathbf{z}) \phi_{fib}(\mathbf{z}) k_1(\mathbf{z}) d\mathbf{z} &\neq 0 \quad \text{for some } \iota
\end{aligned} \tag{G8}$$

The conditions in eq. (G8) are needed to ensure that the transformation induced by $\phi_{fib}(\mathbf{z})$ belong to the defining representation of $\mathcal{G}_{fib}$ on K_D . We can then use the diagonal operators $T(\phi_{fib}, \omega) = \int \phi_{fib}(\mathbf{z}) e^{i\omega(\mathbf{z})} |\mathbf{z}\rangle \langle \mathbf{z}| d\mathbf{z}$, to define congruence transformations belonging to $\mathfrak{G}_{fib}$ as

$$\begin{aligned}
\hat{T}_{fib}(\phi_{fib}, \omega) Y &= T(\phi_{fib}, \omega) Y T(\phi_{fib}, \omega)^\dagger = e^{\check{Y}(\phi_{fib}) + i\check{\omega}} Y e^{\check{Y}(\phi_{fib}) - i\check{\omega}} \\
&= \left(1 + \check{\phi}_{fib}\right)^{\frac{1}{2}} e^{i\check{\omega}} Y \left(1 + \check{\phi}_{fib}\right)^{\frac{1}{2}} e^{-i\check{\omega}} = \left(1 + \hat{\Phi}_{fib}\right) \widehat{e^{i\check{\omega}}} Y
\end{aligned} \tag{G9}$$

where the integral kernel of $\hat{\Phi}_{fibK_D}$ of the projection $\hat{P}_{K_D} \hat{\Phi}_{fib} \hat{P}_{K_D}$ of $\hat{\Phi}_{fibK_D}$ on K_D is defined as $\Phi_{fibK_D}(\mathbf{z}_1, \mathbf{z}_2) = \phi_{fib}(\mathbf{z}_1) \delta(\mathbf{z}_1 - \mathbf{z}_2)$. On diagonal operators $Y_d = Y_{K_D} + \Gamma(\gamma)$ the action of these transformations are

$$\hat{T}_{fib}(\phi_{fib}, \omega) Y_d = \left(1 + \hat{\Phi}_{fib}\right) Y_d = Y_d + \hat{\Phi}_{fibK_D} Y_{K_D} = Y_d + \check{\phi}_{fib} Y_{K_D} \tag{G10}$$

as $\hat{P}_{S_D} \hat{\Phi}_{fib} \hat{P}_{S_D} = \hat{P}_{K_D} \hat{\Phi}_{fib} \hat{P}_{S_D} = 0$. The full intra fiber transformation can thus be expressed as

$$\hat{T}_{fib}(\phi_{fib}, \omega) = e^{\sum_{1 \leq \kappa \leq \infty} \left\{ \Upsilon_{K_{OD}\kappa} | \check{h}_\kappa \right\} (D_{refK_{OD}}^{Nd} | + \Upsilon_{K_D\kappa} | \check{k}_\kappa \right) (D_{refK_D}^{Nd} | \right\}} \tag{G11}$$

where the coefficients $\Upsilon_{K_D\kappa}$ are determined by ϕ_{fib} and ω by the following relationships

$$\begin{aligned}
\mathcal{V}(\phi_{fib})(\mathbf{z}) &= (1 + \phi_{fib}(\mathbf{z}))^{\frac{1}{2}} \\
\Upsilon(\mathbf{z}) &= \mathcal{V}(\phi_{fib})(\mathbf{z}) + i\omega(\mathbf{z}) \\
\Upsilon(\mathbf{z}) &= \sum_{1 \leq \kappa \leq \infty} \Upsilon_{K_D\kappa} \left(|\mathbf{z}\rangle \langle \mathbf{z}| | \check{k}_\kappa \right) (D_{refK_D}^{Nd} | |\mathbf{z}\rangle \langle \mathbf{z}|) \\
\Upsilon_{K_D\kappa} &= \int k_\kappa^{d*}(\mathbf{z}) \Upsilon(\mathbf{z}) D_{refK_D}^N(\mathbf{z}) d\mathbf{z}
\end{aligned} \tag{G12}$$

and the coefficients $\Upsilon_{K_{OD}\kappa}$ are determined by ϕ_{fib} and ω . When acting on D_{ref}^N the transformation $\hat{T}_{fib}(\phi_{fib}, \omega)$ produces all other pure states in $[\gamma]_N$ in a unique fashion and thus

parameterizes the pure states in $[\gamma]_N$ in a non redundant manner in terms of $\{\phi_{fib}, \omega\}$. Note that transformations belonging to $\mathfrak{G}_{fib}$ never change the norm of density operators as they leave $\Gamma(\gamma)$ unchanged.

The transformations that parameterize S_D in a non redundant fashion from a reference pure state D_{ref}^N can always be constructed from selected elements of $\mathfrak{G}_{den}$ acting on Y_{ref} . A complete parameterization of S_D can be achieved in terms of pure states, as given an arbitrary density γ there is *always at least one* pure state that contracts to that density. If we define ϕ_{den} as

$$\begin{aligned}\phi_{den}(\mathbf{z}) &= \sum_{1 \leq \kappa \leq \infty} \phi_{den\kappa}(|\mathbf{z}\rangle \langle \mathbf{z}| |\check{s}_\kappa\rangle) \left(\Gamma(\gamma)^d |\mathbf{z}\rangle \langle \mathbf{z}| \right) \\ \phi_{den\kappa} &= \int s_\kappa^{d*}(\mathbf{z}) \phi_{den}(\mathbf{z}) \Gamma(\gamma)(\mathbf{z}) d\mathbf{z}; \quad 1 \leq \kappa \leq \infty\end{aligned}\tag{G13}$$

where they also satisfy

$$\begin{aligned}\int s_\iota^{d*}(\mathbf{z}) \phi_{den}(\mathbf{z}) s_1(\mathbf{z}) d\mathbf{z} &\neq 0 \quad \text{for some } \iota \\ \int k_\iota^{d*}(\mathbf{z}) \phi_{den}(\mathbf{z}) s_\kappa(\mathbf{z}) d\mathbf{z} &= 0 \quad \forall \iota, \kappa \\ \int s_\iota^{d*}(\mathbf{z}) \phi_{den}(\mathbf{z}) k_\kappa(\mathbf{z}) d\mathbf{z} &= 0 \quad \forall \iota, \kappa \\ \int k_\iota^{d*}(\mathbf{z}) \phi_{den}(\mathbf{z}) k_1(\mathbf{z}) d\mathbf{z} &= 0 \quad \forall \iota, \kappa\end{aligned}\tag{G14}$$

We can use the diagonal operators $T(\phi_{den}) = \int \phi_{den}(\mathbf{z}) |\mathbf{z}\rangle \langle \mathbf{z}| d\mathbf{z}$, to define congruence transformations

$$\begin{aligned}\hat{T}_{den}(\phi_{den}) Y &= T(\phi_{den}) Y T(\phi_{den})^\dagger = e^{\check{Y}(\phi_{den})} Y e^{\check{Y}(\phi_{den})} \\ &= \left(1 + \check{\phi}_{den}\right)^{\frac{1}{2}} Y \left(1 + \check{\phi}_{den}\right)^{\frac{1}{2}} = \left(1 + \hat{\Phi}_{den}\right) Y\end{aligned}\tag{G15}$$

where the integral kernel of $\hat{\Phi}_{denS_D}$ of the projection $\hat{P}_{S_D} \hat{\Phi}_{den} \hat{P}_{S_D}$ of $\hat{\Phi}_{den}$ on S_D is defined as $\Phi_{den}(\mathbf{z}_1, \mathbf{z}_2) = \phi_{den}(\mathbf{z}_1) \delta(\mathbf{z}_1 - \mathbf{z}_2)$. On diagonal operators $Y_d = Y_{K_D} + \Gamma(\gamma)$ this transformation becomes

$$\hat{T}_{den}(\phi_{den}) Y_d = \left(1 + \hat{\Phi}_{den}\right) Y_d = Y_d + \hat{\Phi}_{denS_D} \Gamma(\gamma) = Y_d + \check{\phi}_{den} \Gamma(\gamma)\tag{G16}$$

as $\hat{P}_{K_D} \hat{\Phi}_{den} \hat{P}_{K_D} = \hat{P}_{K_D} \hat{\Phi}_{den} \hat{P}_{S_D} = 0$. The full intra fiber transformation can thus be expressed as

$$\hat{T}_{den}(\phi_{den}, \omega) = e^{\sum_{1 \leq \kappa \leq \infty} \left\{ \Upsilon_{K_{OD}\kappa} | \check{h}_\kappa \right\} \left(\check{Y}_{refK_{OD}}^d | + \Upsilon_{S_D\kappa} | \check{s}_\kappa \right) \left(\Gamma(\gamma)^d | \right) \right\}}\tag{G17}$$

where the coefficients $\Upsilon_{S_D\kappa}$ by the following relationships

$$\begin{aligned}
\mathcal{V}(\phi_{den})(z) &= (1 + \phi_{den}(z))^{\frac{1}{2}} \\
\Upsilon(z) &= \mathcal{V}(\phi_{den})(z) \\
\Upsilon(z) &= \sum_{1 \leq \kappa \leq \infty} \Upsilon_{S_D\kappa}(|z\rangle\langle z| |\check{s}_\kappa) \left(\Gamma(\gamma)^d |z\rangle\langle z| \right) \\
\Upsilon_{K_D\kappa} &= \int \check{s}_\kappa^{d*}(z) \Upsilon(z) \Gamma(\gamma)(z) dz
\end{aligned} \tag{G18}$$

and the coefficients $\Upsilon_{K_{OD}\kappa}$ are determined by ϕ_{den} . When acting on D_{ref}^N the transformation $\hat{T}_{den}(\phi_{den})$ produces a unique set of pure states, one in each equivalence class. Thus $\hat{T}_{den}(\phi_{den})$ acting on the reference pure state D_{ref}^N parameterizes S_D in a non redundant manner in terms of $\{\phi_{den}\}$. It is possible by left multiplying $\hat{T}_{den}(\phi_{den})$ by a transformation $\hat{T}_{fib}(\phi_{fib}(\phi_{den}), \omega(\phi_{den}))$ uniquely determined by $\hat{T}_{den}(\phi_{den})$, that a path of pure reference states $D_{ref}^N(\gamma)$ of a particular type (i.e. Independent Particle, Antisymmetrized Geminal Power etc..) and obtain a more specific non redundant parameterization of S_D .

It is further possible to choose a subset of the transformations given in eq. (G17) that produce normalized pure states, (even from an unnormalized D_{ref}^N), by letting Υ_{S_D1} depend on $\{\Upsilon_{S_D\kappa}; 2 \leq \kappa \leq \infty\}$

APPENDIX H: PRODUCTION OF ARBITRARY PURE STATES FROM A REFERENCE PURE STATE

Here we prove that diagonal transformations, in the sense of Appendix F, can produce any pure state from a reference pure state. The equivalent mathematical statement is that such transformations are transitive on the set of pure states.

Associated with any arbitrarily chosen pure state $|\Psi\rangle \in \mathcal{H}^N = \mathcal{H}^{N_e} \otimes \mathcal{H}^{N_n}$ there is a wavefunction

$$\begin{aligned}
\Psi(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n}) &= \langle \mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n} | \Psi \rangle \\
&= |\Psi(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n})| e^{i\theta_\Psi(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n})} \\
&\equiv e^{\nu_\Psi(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n})} e^{i\theta_\Psi(\mathbf{x}_1, \dots, \mathbf{x}_{N_e} \mathbf{X}_1, \dots, \mathbf{X}_{N_n})}
\end{aligned} \tag{H1}$$

Consider $T_d \in \mathcal{C}(\mathcal{H}^N)$, a diagonal transformation in the sense of Appendix F. Its integral kernel can be written as in eq. (F4) which leads to eq. (F5). The latter equation in

combination with the last RHS of (H1) gives

$$\langle \mathbf{x}, \mathbf{X} | T_d \Psi \rangle = e^{(\mathcal{V}_\Psi + \mathcal{V})} e^{i(\theta_\Psi + \omega)} \quad (\text{H2})$$

with the obvious string of arguments in compressed notation or suppressed for clarity.

If one chooses an arbitrary $|\chi\rangle \in \mathcal{H}^N$, then the choice

$$\begin{aligned} \mathcal{V} &= \mathcal{V}_\chi - \mathcal{V}_\Psi \\ \omega &= \theta_\chi - \theta_\Psi \end{aligned} \quad (\text{H3})$$

generates that $|\chi\rangle$ from the initial $|\Psi\rangle$, itself arbitrarily chosen, via the transformation T_d . Since all physical densities correspond to pure states [?], the result (H3) means that all densities (fibers) can be reached given a pure state corresponding to a reference density (fiber) via transformations of the type developed in Appendix F.

APPENDIX I: PARTITIONED EQUATION OF MOTION IN THE Υ PARAMETERIZATION

This Appendix tracks the reasoning given in Appendix A but for the complex analytic parameterization, $\Upsilon = \mathcal{V} + i\omega$ of the fibers, eq. (55). In this parameterization, the equations of motion that follow from eq. (21) - (23) can be written in matrix partitioned form as

$$\begin{bmatrix} \eta_{\mp\mp}(t) & \eta_{\mp\mp^*}(t) & \eta_{\mp\gamma}(t) \\ \eta_{\mp^*\mp}(t) & \eta_{\mp^*\mp^*}(t) & \eta_{\mp^*\gamma}(t) \\ \eta_{\gamma\mp}(t) & \eta_{\gamma\mp^*}(t) & \eta_{\gamma\gamma}(t) \end{bmatrix} \begin{bmatrix} \mp \\ \mp^* \\ \dot{\gamma} \end{bmatrix} = \begin{bmatrix} \nabla_{\mp}\mathcal{E}(t) \\ \nabla_{\mp^*}\mathcal{E}(t) \\ \nabla_{\gamma}\mathcal{E}(t) \end{bmatrix} \quad (\text{I1})$$

where $*$ denotes complex conjugation. These equations may be written more compactly as

$$\begin{bmatrix} \eta_{FF}(t) & \eta_{F\gamma}(t) \\ \eta_{\gamma F}(t) & \eta_{\gamma\gamma}(t) \end{bmatrix} \begin{bmatrix} \dot{F} \\ \dot{\gamma} \end{bmatrix} = \begin{bmatrix} \nabla_F\mathcal{E}(t) \\ \nabla_{\gamma}\mathcal{E}(t) \end{bmatrix} \quad (\text{I2})$$

with

$$\begin{aligned} F &= \begin{pmatrix} \mp \\ \mp^* \end{pmatrix} ; \eta_{FF}(t) = \begin{pmatrix} \eta_{\mp\mp} & \eta_{\mp\mp^*} \\ \eta_{\mp^*\mp} & \eta_{\mp^*\mp^*} \end{pmatrix} ; \eta_{F\gamma}(t) = \begin{pmatrix} \eta_{\mp^*\gamma} \\ \eta_{\mp\gamma} \end{pmatrix} \\ \eta_{\gamma F}(t) &= \begin{pmatrix} \eta_{\gamma\mp} & \eta_{\gamma\mp^*} \end{pmatrix} ; \nabla_F\mathcal{E}(t) = \begin{pmatrix} \nabla_{\mp}\mathcal{E}(t) \\ \nabla_{\mp^*}\mathcal{E}(t) \end{pmatrix} \end{aligned} \quad (\text{I3})$$

In this F, γ notation the formal structure of the EOM eq. (I2) is exactly as in eq. (A1) of Appendix A with corresponding definitions for S and $\mathcal{E}$. In parallel with the development leading to eq. (A5), it follows that

$$\begin{aligned}\dot{F}(t) &= \eta_{FF}^{-1}(t) [\nabla_F \mathcal{E}(t) - \eta_{F\gamma}(t) \dot{\gamma}] \\ \dot{\gamma}(t) &= [\eta_{\gamma\gamma}(t) - \eta_{\gamma F}(t) \eta_{FF}^{-1}(t) \eta_{F\gamma}(t)]^{-1} \\ &\quad \times [\nabla_\gamma \mathcal{E}(t) - \eta_{\gamma F}(t) \eta_{FF}^{-1}(t) \nabla_F \mathcal{E}(t)]\end{aligned}\tag{I4}$$

Under the assumption that $\eta_{FF}^{-1}(t)$ exists, denote the solution of the first of these for the correct density $\gamma_\#(t)$ by $F_\#(t)$. Then the exact equation of motion for the temporal development of the density is

$$\dot{\gamma}(t) = \mathcal{N}_1(\gamma(t)) [\nabla_\gamma \mathcal{E}(\gamma(t)) - \mathcal{N}_2(\gamma(t))]\tag{I5}$$

The functionals are defined as

$$\begin{aligned}\mathcal{N}_1(\gamma(t)) &= [\eta_{\gamma\gamma}(\zeta_\#(t), \gamma(t)) - \eta_{\gamma F}(\zeta_\#(t), \gamma(t)) \eta_{FF}^{-1}(\zeta_\#(t), \gamma(t)) \times \\ &\quad \eta_{F\gamma}(\zeta_\#(t), \gamma(t))]\^{-1}\end{aligned}\tag{I6}$$

$$\mathcal{N}_2(\gamma(t)) = \eta_{\gamma F}(\zeta_\#(t), \gamma(t)) \eta_{FF}^{-1}(\zeta_\#(t), \gamma(t)) \nabla_F \mathcal{E}(\gamma(t)).\tag{I7}$$

$$\mathcal{E}(\gamma(t)) = \frac{\langle \Psi(\zeta_\#(t), \gamma(t)) | H | \Psi(\zeta_\#(t), \gamma(t)) \rangle}{\langle \Psi(\zeta_\#(t), \gamma(t)) | \Psi(\zeta_\#(t), \gamma(t)) \rangle}\tag{I8}$$

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